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Disentangling Sequential Goal Representations in A Maze-solving Agent

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Declaration

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Abstract

Reinforcement learning (RL) agents trained on multi-objective tasks must track and switch between sequential goals to succeed. While such agents can learn complex goal-directed behaviour, how neural networks internally represent and select between multiple similar objectives remains poorly understood. In this work, we show that goal selection in a maze-solving RL agent is mediated by a spatial gating mechanism where negative activations mark regions of interest, with these patterns shifting systematically as objectives are completed. Clamping experiments confirm this operates spatially: the agent avoids any clamped region regardless of which entity occupies it. We further identify a two-stage processing pipeline where early convolutional layers establish the target entity before fully-connected layers process directional information. This mechanism was discovered through simple mean activation analysis rather than sophisticated interpretability tools such as SAEs, demonstrating that approaching interpretability for reinforcement learning might require different considerations than is typically used for other machine learning domains. Our findings provide a concrete mechanistic account of goal representation in this RL agent. More broadly, spatial gating via negative activations may be a general strategy for specifying regions of interest in complex RL domains.

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Chapter 1

Introduction

Reinforcement learning agents are increasingly deployed in critical infrastructure, from data centres (Luo et al., 2022) to electrical grids (Nakabi and Toivanen, 2021) and nuclear fusion reactors (Degraeve et al., 2022). Yet how these agents internally represent and pursue goals remains poorly understood. This opacity poses significant risks: without understanding goal representations, we cannot ensure alignment, debug failures, or build reliable autonomous systems (Amodei et al., 2016).

While mechanistic interpretability (MI) has made substantial progress in understanding Large Language Models (Elhage, Nanda, et al., 2021; McDougall et al., 2023; Olsson et al., 2022; Wang et al., 2022), RL agents have received comparatively little attention, with notable exceptions (Atrey et al., 2020; Hilton et al., 2020). This gap is significant: RL outputs are action sequences rather than words or images, making interpretation inherently more challenging since actions are often only meaningful over broader sequences. Yet RL environments offer a unique opportunity for MI research, as they contain well-defined goals and clear failure modes that allow direct testing of mechanistic hypotheses. Indeed, recent work on steering LLMs (Turner et al., 2024a) was directly inspired by interpretability work on maze-solving agents (Mini et al., 2023).

This thesis investigates how neural networks represent and pursue multiple sequential goals in RL agents. Building on (Mini et al., 2023), which identified channels controlling goal pursuit in single-objective mazes, we extend this analysis to the Procgen Heist environment where agents must collect up to three keys in sequence before reaching a gem (Cobbe, Hesse, et al., 2020). A key challenge in this investigation is polysemanticity (Elhage, Hume, et al., 2022), the phenomenon where individual neurons respond to multiple distinct features, which can obscure goal representations and complicate interpretation.

1.1 Background and Motivation

RL is a field of machine learning that has the unique promise of delivering super-human performance in a wide range of environments (Silver et al., 2016). However, the black box nature of these models makes it difficult to understand how they achieve these results.

RL agents learn through interaction with an environment, receiving observations and rewards that guide policy development (R. S. Sutton and A. G. Barto, 2018). The field of MI seeks to decode the black boxes of neural networks so that these models can be used with greater confidence and reliability (Olah, Satyanarayan, et al., 2018). Recent work provides compelling evidence that neural networks can learn meaningful internal models and conceptual abstractions from data (K. Li, Aspen K Hopkins, et al., 2022), suggesting that understanding goal representations in one model may inform investigations of others.

However, not all goal-seeking models are suitable for this research; some tasks are simple enough to solve without developing complex mechanisms to represent abstract goals. The Procgen environments stand out as good environments for this reason, as they procedurally generate environments for the agent to solve. This means that the agent cannot simply overfit to the data, but has to learn general patterns in order to score well on the environment, which necessitates developing a complex representation of goals within the model.

We chose Procgen Heist in particular to explore how multiple goals might be represented in the model, as previous work has already shown the presence of ‘cheese representations’ with the Procgen Maze environment (Mini et al., 2023). Because advanced AI agents would be operating in environments with multiple goals, the question of whether there was a substantial difference in the representation of these goals is an essential one.

However, as the environment becomes more complex, the model does too, and polysemanticity becomes a significant hurdle to understanding the model’s operations. Specifically, whether the model’s representations of goals represent a ‘true’ representation or simply result from polysemanticity or pattern memorisation in the network. This led to the use of Sparse Autoencoders (SAEs), as well as a variety of fundamental interpretability approaches, which are discussed more in Section 2.2, to confirm whether channels in the network naturally tend to represent the different entities using dedicated channels, or whether there are single ‘current goal’ channels in the network that are re-used across different entities.

1.2 Problem Statement

How do RL agents internally represent and select between multiple sequential goals? Without understanding this, we cannot verify that agents pursue intended

objectives, diagnose failures when they occur, or predict behaviour in novel situations. This problem is particularly acute in multi-objective environments where the same visual features (keys, doors, targets) must be processed differently depending on task progress. The inherent opacity of deep neural networks, compounded by polysemanticity, makes isolating goal-related representations challenging. This thesis addresses this gap by investigating goal representation mechanisms in the Procgen Heist environment, where agents must track and switch between multiple sequential objectives.

1.3 Research Questions

This research seeks to determine the key functional components in a maze-solving agent that are responsible for determining the agent’s goal, and to provide clearer evidence of how the network learns, represents, and pursues sequential goals. To this end, we investigate the following key questions:

- RQ1:** How are the different goals encoded in the network?
- RQ2:** What mechanism does the model use to determine which entity to target?
- RQ3:** Can we use SAEs to better determine the role of specific features?
- RQ4:** Do specific channels target specific entities, or are there general goal steering channels in the network?
- RQ5:** Why do we observe different rates of success when steering the agent away from different entities using activation patching?
- RQ6:** Do different neural network layers play specialised roles in how the agent processes goal information or becomes steerable?

1.4 Key Hypotheses

Based on the preceding research questions, this study investigates the following key hypotheses:

- H1:** We hypothesise that the model predominantly learns to represent specific goals (like individual keys or the gem) via dedicated channels, rather than relying solely on generalised ‘current target’ signals or highly distributed, non-specific encodings (Bau et al., 2017; Cammarata et al., 2020).
- H2:** We hypothesise that Sparse Autoencoders (SAEs) deconstruct complex, polysemantic activations within the agent’s layers into a more disentangled and interpretable set of features, thereby clarifying the specific roles of channels into specific circuits related to goals (Bricken et al., 2023).

- H3:** We hypothesise that the model has a strong hierarchical preference structure that is encoded deeply in the weights (Langosco et al., 2022).
- H4:** We hypothesise that SAEs can be trained to reconstruct the original model’s activations with sufficient fidelity to preserve the agent’s task performance, indicating that the features learned by the SAE are indeed relevant to the agent’s operational policy (Bricken et al., 2023; Templeton et al., 2024).
- H5:** We hypothesise that distinct layers within the Procgen agent’s network play specialised roles in processing goal-related information, potentially with earlier layers focusing on perceptual parsing of the location of entities and the structure of the maze, while intermediate layers encode more abstract information such as which entity to pursue and while later layers produce plans for how to reach it (Bau et al., 2017; Trim and Grayston, 2024; Zeiler and Fergus, 2013).

1.5 Scope and Limitations

We focus on a single environment (Procgen Heist) to perform deep mechanistic analysis rather than a broad empirical analysis. This approach follows established precedent in RL interpretability research (Hilton et al., 2020; Mini et al., 2023; Trim and Grayston, 2024), where thorough investigation of specific mechanisms within one environment has proven more valuable than superficial analysis across many domains. The mechanisms we discover raise questions worth investigating in other multi-objective RL domains, though we cannot assume they generalise beyond this specific setting.

We trained a custom model from scratch rather than using a typical variant for this environment, using the architecture described in (Hilton et al., 2020). This simplified model allowed for analysis of a narrower surface area, by using fewer layers and removing skip connections that otherwise confound interpretation.

Additionally, the nature of the RL environment itself poses a challenge; much of the model’s complexity is likely encoded in functionally critical but semantically uninterpretable factors, such as subtle patterns for navigating obstacles, and is further complicated by the fact that the network has a split policy and value head, meaning some of the work done in the later layers is not entirely to do with actions but also in calculating the value of the current state, the impact of which is challenging to measure. While SAEs and probes may successfully isolate high-level goal features, they may struggle to render these low-level navigational strategies intelligible. Consequently, interpreting the full suite of discovered features remains a significant challenge, as our tools may not capture all of the original model’s computational strategies.

1.6 Significance of the Study

This work applies SAEs to an RL environment, which remains relatively uncommon. A great deal of recent MI work has been applied to LLMs rather than RL agents, meaning that the space of tools and approaches is relatively small compared to other types of neural networks. Our hope is that in the extensive application of these tools to this network we can help to mature the discourse on approaching these sorts of problem and clarify the strengths and weaknesses of these tools in this domain.

We discover that the network uses spatial gating through the use of negative and positive activations in the CNN layers, where negative activations mark areas of the maze that the agent still needs to go to, and which are filled in with positive activations as entities in those regions are collected. This finding has implications for how different entities within an environment are captured by a model trained with RL. Some of the experimental frameworks used provide a replicable template for future studies seeking to causally understand goal-directed behaviours in neural networks.

1.7 Thesis Structure

This thesis is structured as follows: Chapter 2 provides a review of the relevant literature in RL and MI. Chapter 3 details the experimental setup, including the environment, model architecture, and the interpretability techniques employed. Chapter 4 presents the findings from the experiments, including activation visualisations, probing results, parallel rollout analysis, and steering experiments. Chapter 5 discusses the implications of these findings, and Chapter 6 concludes the thesis and suggests avenues for future research.

Chapter 2

Literature Review

2.1 Introduction

This chapter reviews the foundational literature underpinning this thesis, covering reinforcement learning fundamentals, mechanistic interpretability techniques, and their intersection in RL agent analysis. We examine how prior work has approached understanding goal representations in neural networks, and identify the gaps this thesis addresses.

2.1.1 Reinforcement Learning Fundamentals

This section provides a brief overview of the fundamental concepts in RL that are relevant to the research discussed. While a baseline understanding of RL and AI concepts is assumed, key elements of the environments and training paradigms employed from the literature are reviewed.

At its core, the problem that RL solves can be formalised as a Markov Decision Process (MDP), or Partially Observable Markov Decision Process (POMDP). An MDP is a mathematical framework for modelling decision-making in situations where the outcomes of actions may not be predictable to the agent. Environments can be both entirely deterministic or stochastic in nature. MDPs are defined by a tuple (S, A, P, R, γ) , representing the set of states S , the set of actions A , the state transition probability function $P(s'|s, a)$, the reward function $R(s, a, s')$, and a discount factor $\gamma \in [0, 1]$ that balances immediate and future rewards. The transition function $P(s'|s, a)$ gives the probability of transitioning to next state s' given current state s and action a , where the notation $s'|s$ denotes the conditional probability of the next state given the current state. The agent's goal is to learn a strategy that maximises the cumulative discounted reward over time.

Markov Decision Processes

The objective of the training process is formalised as an optimisation process maximising expected discounted return:

$$J(\pi) = \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t r_t \right]$$

where π is the policy, τ denotes a trajectory sampled from following policy π , and r_t is the reward received at time step t .

Value Functions and Policies

The process of training a model involves iterating through many interactions with the environment to produce a policy denoted as $\pi(a|s)$, which is a mapping from states to a probability distribution over actions. The goal of the RL training process is to find an optimal policy, π^* , that maximises the agent's expected reward over time. RL often makes use of value functions (R. S. Sutton and A. G. Barto, 2018) to estimate expected returns. The state-value function, $V^\pi(s)$, represents the expected return following policy π given state s , and the action-value function, $Q^\pi(s, a)$, corresponds to the expected return from continuing to follow policy π after taking action a in state s .

Value functions are foundational to many training algorithms. Two fundamental approaches exist for estimating them: Monte Carlo methods, which use complete episode returns, and Temporal Difference (TD) learning, which bootstraps from current estimates. These algorithms alternate between sampling data from the environment and optimising objectives via stochastic gradient ascent.

Bellman Optimality Equations

RL is fundamentally about finding the optimal policy π^* that maximises expected returns. This problem can be characterised through the Bellman optimality equations, which provide a recursive decomposition of optimal value according to observations of states:

$$V^*(s) = \max_a \sum_{s'} P(s'|s, a) [R(s, a, s') + \gamma V^*(s')] \quad (2.1)$$

$$Q^*(s, a) = \sum_{s'} P(s'|s, a) [R(s, a, s') + \gamma \max_{a'} Q^*(s', a')] \quad (2.2)$$

These equations express the optimal value of a state as the maximum expected sum of immediate reward plus the discounted value of optimal successor states. $V^*(s)$ represents the optimal value of being in state s , while $\max_a$ describes the selection

of the action that yields the highest expected return; $P(s'|s, a)$ is the probability of transitioning to state s' when taking action a in state s , $R(s, a, s')$ is the reward received immediately from this transition, and $\gamma V^*(s')$ is the discounted optimal value of the next state. Similarly, $Q^*(s, a)$ represents the optimal value of taking action a in state s , where $\max_{a'}$ selects the best subsequent action in the next state.

This allows the recursive decomposition of optimal decision making into taking the best known action currently available, conditioned on continuing to follow the optimal policy thereafter. The Bellman equations form the theoretical foundation for many RL algorithms, providing both a characterisation of optimality and a computational framework for achieving it. In settings with small, discrete state spaces, these equations can be solved directly through dynamic programming methods such as value iteration or policy iteration (R. S. Sutton and A. G. Barto, 2018). However, as we discuss in the following section, the computational demands of modern RL environments necessitate approximation approaches that trade theoretical guarantees for practical scalability.

From Tabular Methods to Function Approximation

Early RL methods operated in discrete state spaces where optimal value functions could be represented precisely in lookup tables. For such tabular methods, convergence to the optimal policy is guaranteed when using algorithms based on the Bellman optimality equations. In this setting, each state-action pair (s, a) has an explicit entry in a table storing its Q-value, and the Bellman equations can be applied iteratively until convergence.

However, visual environments such as Atari games have state spaces that would be computationally infeasible to solve with a lookup table ($64 \times 64 \times 3 = 12,288$ dimensional input with $256^{12,288}$ possible states for 8-bit RGB images) (Mnih, Kavukcuoglu, Silver, Graves, et al., 2013). This necessitates function approximation using neural networks to approximate the value functions, specifically approximating either the state-value function $V(s)$ or the action-value function $Q(s, a)$. Rather than storing explicit values for every state or state-action pair, a parameterised function $V_\theta(s)$ or $Q_\theta(s, a)$ learns to generalise across similar states, enabling the agent to estimate values for states never previously encountered, and allowing for remarkable generalisation Mnih, Kavukcuoglu, Silver, Rusu, et al., 2015.

Function approximation does not provide the convergence guarantees of tabular methods, and causes the learned representations in our models to become uninterpretable due to the arbitrary decomposition of features and things like polysemanticity, which will be discussed in Section 2.2.2.

Temporal Difference Learning

There are two fundamental approaches to estimating value functions. Monte Carlo methods average the actual returns observed after visiting each state across complete episodes. While conceptually simple and unbiased, these methods suffer from high variance (since returns depend on many stochastic transitions) and cannot learn online; they must wait until an episode terminates to update value estimates.

Temporal Difference (TD) learning takes a different approach, updating value estimates after each transition by bootstrapping from current estimates rather than waiting for complete episode returns (R. S. Sutton and A. G. Barto, 2018). The TD(0) update rule is given by:

$$V(s_t) \leftarrow V(s_t) + \alpha[r_t + \gamma V(s_{t+1}) - V(s_t)]$$

where α denotes the learning rate. The term $\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$ is the TD error, representing the discrepancy between the current value estimate and the improved estimate formed by the immediate reward and the discounted value of the successor state. This mechanism allows the agent to learn from each transition. Unlike Monte Carlo methods, which require complete episodes and are unbiased but high-variance, TD methods introduce bias by the process of bootstrapping but significantly reduce variance and allow for online learning. TD error serves as the primary learning signal for value-based and actor-critic algorithms discussed in the following sections.

2.1.2 Development of Deep RL

Value-Based Deep RL

Value-based methods learn the value of states or state-action pairs and derive a policy by selecting actions that maximise expected value. Rather than directly learning which action to take, these methods estimate how valuable each action is in each state, then act greedily with respect to these estimates.

Deep Q-Networks (DQNs) (Mnih, Kavukcuoglu, Silver, Rusu, et al., 2015) achieved a landmark breakthrough by demonstrating that a single neural network architecture, trained with a unified algorithm and hyperparameter configuration, could achieve human-level performance across 29 out of 49 diverse classical Atari games (games built for arcades that include repetitive patterns and structures). This represented a significant advance over prior approaches that required game-specific feature engineering or algorithm tuning, showing that deep learning combined with RL could learn general-purpose representations directly from raw pixel inputs.

The key innovations they contributed were:

Experience replay buffers that store state transitions (s, a, r, s') and are sampled uniformly at training time. This breaks correlation between observations that

would cause the model to ‘chase its own tail’ during training, due to the model getting into a feedback loop that causes it to overly sample a given region of the state space. Sampling from the replay buffer allows the network to learn from the same observation repeatedly and prevents catastrophic forgetting.

The use of target networks Q_{θ^-} that are periodically copied from the main network Q_{θ} that provide stable targets for TD learning. While the main network Q_{θ} updates every gradient step, the target Q_{θ^-} is overwritten by the main network periodically (typically every 10,000 steps). This avoids the problem of constantly chasing a moving target, by freezing the optimisation target allowing the opportunity to make steady progress towards a stable optimisation target, until that target is updated again.

These contributions provided sufficient stability for RL training to meet and surpass human performance on challenging Atari games, proving that convolutional neural networks (CNNs), which use filters trained to process information in grid structures, allow Deep RL networks to learn meaningful features from pixels. However, value-based methods struggle in continuous action spaces, where computing the argmax over an infinite number of possible actions becomes intractable. This limitation led to increased focus on policy gradient methods, which can naturally handle continuous actions by directly parameterising a policy distribution.

Policy Gradient Methods

Policy gradient methods arise from the natural challenges introduced by trying to learn optimal policies within continuous action spaces and with non-deterministic policies. Rather than learning state values using these to indirectly develop a policy, policy gradients directly optimise parameterised policy $\pi_{\theta}(a|s)$ by gradient ascent on expected returns.

The policy gradient theorem (R. S. Sutton, McAllester, et al., 1999) provides the insight required for this approach. For an objective $J(\theta) = \mathbb{E}_{\tau \sim \pi_{\theta}}[R(\tau)]$, the gradient can be computed without requiring pre-existing knowledge of the environment:

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\pi_{\theta}} \left[\sum_{t=0}^T \nabla_{\theta} \log \pi_{\theta}(a_t|s_t) \cdot G_t \right]$$

where $G_t = \sum_{k=t}^T \gamma^{k-t} r_k$ is the discounted return from timestep t . This approach allows the development of a policy requiring only the ability to sample trajectories and differentiate the policy network without any model of environment dynamics.

However, vanilla policy gradient methods suffer from erratic training dynamics because returns can vary dramatically between episodes. Actor-critic methods use advantage estimates to ensure that model updates are in a consistent range around zero with the function $A(s, a) = Q(s, a) - V(s)$, instead of directly updating the policy on expected returns. The advantage measures how much better an action is

compared to the expected value from the current state. This fixes the final updates to the network around zero, reducing variance without altering the direction of the gradients (Schulman, Moritz, et al., 2018).

Actor-Critic methods

Actor-critic methods combine useful aspects of both policy and value based methods by maintaining both a policy network (actor) and a value function (critic). The actor produces actions (or probability distributions over actions) and the critic estimates the quality of the actions by estimating the value of the proceeding state. This provides a stable signal of progress (learned baseline) from the critic that reduces variance without adding bias (R. Sutton and A. Barto, 1998).

The core actor-critic update uses the TD error as an unbiased estimate of the advantage:

$$\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$$

The actor updates using:

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\pi_{\theta}}[\nabla_{\theta} \log \pi_{\theta}(a_t|s_t) \cdot \delta_t]$$

While the critic updates to minimise:

$$L_V = \mathbb{E}_{\pi_{\theta}}[(r_t + \gamma V(s_{t+1}) - V(s_t))^2]$$

The actor-critic approach addresses critical limitations in policy gradient methods. The critic’s value estimates replace high variance Monte Carlo reward estimates and provide lower variance targets, providing more stable learning. The actor can also learn stochastic policies, due to being able to produce probability distributions, rather than being constrained to the estimates of a purely deterministic value function.

Modern implementations such as A3C and A2C (Mnih, Badia, et al., 2016) typically share convolutional encoders between actor and critic for computational efficiency, though this creates potential entanglements in model representations that can provide challenges for interpretability.

Generalised Advantage Estimation

Generalised Advantage Estimation (GAE) (Schulman, Moritz, et al., 2018) addresses a key trade-off between having high variances or low bias updates in actor-critic training updates. It acts as a method to interpolate between high variance and low bias (Monte Carlo methods), or methods that are low variance but high bias (single-step TD error). GAE allows for trading off along this spectrum using exponentially-weighted averaging of n-step advantage estimates.

A mathematical formulation of the GAE estimator:

$$\hat{A}_t^{\{\text{GAE}(\gamma, \lambda)\}} = \sum_{l=0}^{\infty} (\gamma \lambda)^l \delta_{t+l}^V$$

where $\delta_t^V = r_t + \gamma V(s_{t+1}) - V(s_t)$ is the temporal difference (TD) residual. This can be interpreted as a discounted sum of TD errors, or in other words, as an exponentially-weighted average of all n-step advantage estimates. The use of a hyperparameter $\lambda \in [0, 1]$ controls the bias-variance trade-off. When $\lambda = 0$, GAE reduces to the one-step TD error (high bias, low variance), while $\lambda = 1$ produces Monte Carlo returns minus baseline (low bias, high variance).

In practice, $\lambda \approx 0.95$ is typically used (Schulman, Moritz, et al., 2018), using information from around 20 steps into the future while providing meaningful variance reduction. For sparse reward environments, GAE enables credit to backward propagate through the value function while preventing potentially disruptive variance that may arise from Monte-Carlo methods.

Proximal Policy Optimization

PPO is a policy gradient based approach that aims to achieve stable policy updates that are large enough to make meaningful progress but constrained enough to avoid destabilising the training process.

The approach is inspired by Trust Region Policy Optimization (TRPO) (Schulman, Levine, et al., 2017), which constrains policy updates to a “trust region” (a region in parameter space where a local approximation to the objective remains valid). TRPO enforces this constraint by solving a constrained optimisation problem that limits the KL divergence between old and new policies. This requires computing natural gradient updates, which involve approximating the inverse Fisher information matrix using conjugate gradient methods; a computationally expensive second-order optimisation.

PPO achieves similar stability to TRPO using only first-order gradients: rather than computing curvature information (how far you can safely step without overshooting) via the Fisher matrix, PPO clips the objective function to discourage large policy changes. The clipped surrogate objective is:

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t [\min(r_t(\theta)A_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)A_t)]$$

where $\mathbb{E}_t$ denotes the empirical average over a finite batch of samples collected during a rollout, and $r_t(\theta) = \pi_\theta(a_t|s_t)/\pi_{\theta_{\text{old}}}(a_t|s_t)$ represents the probability ratio between new and old policies. This ratio is clipped to $[1 - \epsilon, 1 + \epsilon]$ (typically with $\epsilon = 0.2$), which means that samples whose probability ratios exceed the clipping range contribute zero gradient to the update step. This prevents individual experiences

with large advantages (A_t) from changing the policy drastically while still allowing learning from other samples in the batch.

PPO typically employs Generalized Advantage Estimation (GAE, discussed in Section 2.1.2) to compute the advantage terms A_t , allowing for modifiable bias-variance tradeoff that is particularly helpful for environments that have sparse rewards. Algorithm 1 summarises PPO, where θ denotes the policy parameters, ϕ denotes the value function parameters, V_ϕ is the learned value function, and $\hat{R}_t = \sum_{k=t}^T \gamma^{k-t} r_k$ is the discounted return from timestep t .

Algorithm 1: Proximal Policy Optimization (PPO)

Input: Initial policy parameters θ_0 , initial value function parameters ϕ_0

Input: Clipping threshold ϵ , number of epochs K , minibatch size M

for $iteration = 1, 2, \dots$ **do**

 Collect trajectories $\mathcal{D} = \{\tau_i\}$ by running policy π_θ in environment;

 Compute rewards-to-go $\hat{R}_t$ and advantage estimates $\hat{A}_t$ using GAE;

 Store old policy probabilities $\pi_{\theta_{\text{old}}}(a_t|s_t)$ for all $(s_t, a_t) \in \mathcal{D}$;

for $epoch = 1, \dots, K$ **do**

for minibatch $\mathcal{B} \subset \mathcal{D}$ **do**

 Compute probability ratio: $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$;

 Compute clipped surrogate objective::

$$L^{\text{CLIP}} = \frac{1}{|\mathcal{B}|} \sum_{t \in \mathcal{B}} \min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon) \hat{A}_t \right);$$

 Compute value loss: $L^V = \frac{1}{|\mathcal{B}|} \sum_{t \in \mathcal{B}} (V_\phi(s_t) - \hat{R}_t)^2$;

 Update θ by gradient ascent on L^{CLIP} ;

 Update ϕ by gradient descent on L^V ;

2.1.3 Environments

Sparse Rewards and Credit Assignment

Environments with sparse rewards pose fundamental challenges for RL training. In such environments, effective actions and the final reward may be separated by a large number of steps. This is known as the credit assignment problem (R. S. Sutton, McAllester, et al., 1999). With typical discount factors of $\gamma = 0.99$, early actions in a 500-step episode receive gradient signals approximately $0.99^{500} \approx 0.007$ times weaker than actions near the conclusion of an episode. This means that early game strategies cannot be learned straightforwardly by direct reinforcement.

The challenge is worsened by the enormous number of possible action sequences. In a 300-step episode with 15 possible actions, there exist 15^{300} potential trajectories. Because the chances of success are so low with a random walk, learning from early training at all can be extremely challenging, and what little signal there is, is hard

to learn from.

Actor-critic architectures with GAE (Section 2.1.2) mitigate this challenge, though they do not eliminate it entirely. The critic performs credit assignment via temporal difference learning, propagating value estimates backward step-by-step rather than waiting for terminal rewards as direct policy gradient methods must. Once the critic has learned from initial successes, it provides a value estimate for every state visited, enabling the actor to learn from advantage estimates at every timestep rather than only when sparse rewards are received. However, this bootstrapping process still requires some initial successful episodes to provide signal. Once initial learning has occurred, PPO (Section 2.1.2) outperforms other methods by ensuring that these learned value estimates translate into stable policy improvements, preventing destructive updates that could slow down or stop training progress.

The Procgen Benchmark

The Procgen benchmark (Cobbe, Hesse, et al., 2020) consists of 16 procedurally generated environments based on Atari style games. These procedurally generated environments prevent easily memorised solutions and force agents to learn general strategies and representations of the environment.

An example of a maze in Procgen Heist is shown in Figure 2.1. The structure of the environment is as follows:

- **States:** 64×64 RGB pixel observations of the maze
- **Actions:** Discrete movement commands (vertical, horizontal and diagonal movements)
- **Rewards:** Sparse. +10 for reaching the gem, 0 otherwise
- **Transitions:** Deterministic maze physics

Because the agent receives the entire $64 \times 64 \times 3$ (height by width by red, blue and green colour channels) pixel state space of the maze as its observation, the Procgen Heist environment is a fully-observable MDP, where the state is directly represented by the agent’s visual input.

The basic problem for the agent in Heist is to navigate to a series of objects in the environment, each necessary to advance. The keys must be collected to open the locks, at which point the locks disappear. When a key is collected it is also visibly displayed in the top right corner, so that information about the state is preserved. The maze is solved when the agent reaches the gem. Heist’s design includes a natural curriculum: episodes randomly generate 0 to 3 keys, allowing agents to first master simple gem-only mazes before tackling multi-key sequences, though the distribution of maze types is sampled uniformly during training. This

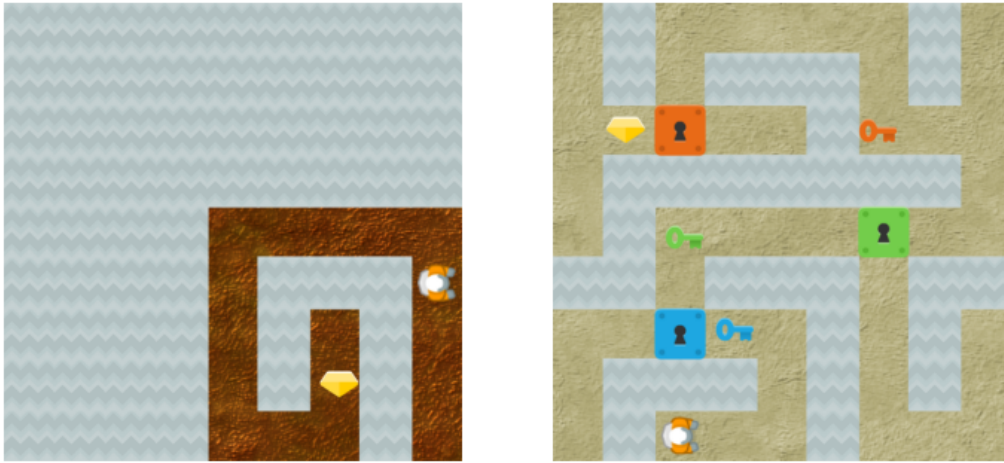


Figure 2.1: The Procgen Heist environment presents a procedurally generated maze where agents must collect keys and unlock corresponding doors in a specific sequence (blue, green, red) before reaching the final goal (gem). The environment varies in complexity, sometimes only having the gem present, and other times including multiple keys and locks.

curriculum provides crucial scaffolding, as agents can learn basic navigation on easy layouts while gradually building representations for more complex mazes.

2.1.4 Architectures

CNN Architectures for Visual RL

While the algorithms discussed above are general-purpose, applying them to visual environments requires neural network architectures capable of processing high-dimensional image inputs. Initial DQN implementations used a simple 3-layer CNN, sufficient for Atari’s 84×84 pixel inputs (Mnih, Kavukcuoglu, Silver, Rusu, et al., 2015). However, more complex visual environments such as Procgen (Cobbe, Hesse, et al., 2020) demanded deeper architectures with new methods for scaling model capacity, including residual connections (He et al., 2015) and the IMPALA encoder (Espeholt et al., 2018).

A key technique enabling deeper networks is residual connections (He et al., 2015). Standard deep networks suffer from vanishing gradients, where training signal weakens as it propagates backward through many layers. Residual connections address this by adding skip connections that allow gradients to flow directly through the network. Each residual block learns a function $F(x)$ that is added to its input, so the output is $x + F(x)$. If the optimal transformation is close to the identity of the inputs, the network can simply learn $F(x) \approx 0$ (He et al., 2016). This approach

allowed for the training of deeper (15+) layer networks.

To improve computational efficiency, most modern actor-critic implementations share a single encoder (the layers that process raw input into a learned representation) between the policy and value networks. Separate output heads then produce actions and value estimates (Espeholt et al., 2018; Mnih, Badia, et al., 2016; Schulman, Wolski, et al., 2017). For visual RL, this encoder typically consists of convolutional layers. This parameter sharing enhances sample efficiency by allowing both objectives to benefit from shared feature learning. However, this creates a multi-task learning problem where features must simultaneously encode ‘what action to take’ and ‘how valuable is this state’, potentially conflicting objectives that can lead to entangled representations. (Hilton et al., 2020) demonstrated this challenge in Procgen Coinrun agents, showing that individual channels often respond to multiple unrelated game elements, making it difficult to pinpoint the function of specific network components.

IMPALA Models

Among the various architectures developed for visual RL, the IMPALA encoder has emerged as a particularly effective baseline for challenging environments like Procgen. Originally introduced by (Espeholt et al., 2018) as a scalable RL framework, its convolutional encoder has since become a primary baseline for Procgen and Atari agents (Espeholt et al., 2018). The encoder consists of three “IMPALA blocks”, each containing two 3×3 convolutions with ReLU activations followed by 2×2 max-pooling that halves spatial resolution. Every block doubles the number of channels (independent 2D feature maps, where each channel’s convolutional filter learns to detect a different pattern in the input), progressing from $16 \rightarrow 32 \rightarrow 64 \rightarrow 128$ in the original IMPALA architecture, though in work such as that performed by (Mini et al., 2023) the channel count remains fixed at 16. A final fully-connected layer produces a 256-dimensional embedding that feeds into two heads: a policy head outputting action probabilities and a value head producing state value estimates.

2.2 An Overview of Modern ML Interpretability

Mechanistic interpretability (MI) aims to understand neural network behaviour by analysing internal structure and intermediate computations, rather than treating models as black boxes. This section reviews the key techniques and challenges in the field.

2.2.1 Interpretability as a Safety Solution

MI as a field aims to provide a precise, granular, and above all mechanistic explanation of the external behaviour of neural networks by analysing their internal structure and intermediate computations. This enables us to understand why a model produced a certain output, using tools applied to the model itself, and not simply to its inputs and outputs. Within the risk framework discussed in Section 2.4.1, MI is useful primarily in the scope of vulnerability, in that it gives us tools that make it less likely for a disaster to occur. We generally use MI as a post hoc intervention, once the model has already been trained, though recent work also uses MI tools to shape training in ways that give us greater control (Casademunt et al., 2025). These interventions improve our ability to monitor what the model is doing beyond simply observing its outputs, such as by using probes (linear classifiers trained on internal activations to detect specific features; see Section 2.2.6) to identify potentially harmful or deceptive outputs before they are generated.

As AI systems approach and exceed human capabilities, and the safety concerns around their deployment grow, MI offers the potential to confirm that models are being faithful in their outputs and are acting in ways that we can verify are safe and appropriate. In RL domains, this is particularly the case in high risk deployment scenarios, or in cases where there may be a significant shift in distribution, and behaviour verification ahead of time is challenging. If we have insight into key mechanisms within the model, we can predict and prevent potentially harmful behaviours before they happen.

One attractive quality of MI is that it offers tools which are powerful while also being cheap to implement and effective at providing insight into model operations. Linear probes (Alain and Bengio, 2018), for example, can be trained with trivially small amounts of compute to detect what goals a model is pursuing or whether it is engaging in reward hacking (Shihab et al., 2025). This makes MI quite promising for organisations involved in the deployment of frontier LLMs, where the costs associated with training models can be extreme.

The promise of MI is already being observed in applications where it was either difficult or previously impossible to make progress. One study by (Marks et al., 2025) involved a set up of a single red team and multiple blue teams, where the red team trained a model organism (a model modified to elicit a specific behaviour) that had a hidden goal. The blue teams were equipped with different sets of tools, and directed to identify the hidden goal in the model organism within a limited time. The researchers found that the team with SAE access discovered the hidden objective in 90 minutes, compared to approximately 6 hours for teams without SAEs, while the team with only black-box API access never discovered it despite over 70 researcher-hours of effort.

In another example, researchers used MI techniques to extract a novel chess con-

cept from AlphaZero and successfully taught it to grandmasters, who reported genuinely learning something new about the game. (Cunningham et al., 2025) demonstrated that linear probes trained on intermediate activations can detect jailbreak attempts and dangerous content (including bioweapon-related queries) in real-time, achieving detection performance comparable to dedicated classifiers with orders of magnitude more parameters.

In spite of the above, there are likely real limits to what we can expect from the field of MI. For example, it seems unlikely that the complete reverse engineering of frontier AI systems is attainable in the near future. We might only be able to achieve a precise understanding of a small part of a neural network’s functioning with significant effort. Even this may be enough, if the particular feature we are interested in is sufficiently critical.

2.2.2 Superposition

One significant challenge that arises when doing MI is that representations of concepts within neurons are often not clean singular representations, but instead may be smeared across multiple neurons or have multiple representations encoded within a single neuron. The latter case, polysemanticity, was first publicly observed in the ‘curve-circuits’ analyses of Inception V1, where single units responded to several distinct curve orientations (Cammarata et al., 2020).

This phenomenon is now understood as a consequence of superposition. Because deep layers have limited dimensionality, networks compress many latent directions into overlapping linear combinations, so a neuron’s activation mixes several underlying features (Elhage, Hume, et al., 2022; Olah, Cammarata, et al., 2020). This typically happens with features that are unlikely to fire under the same input distribution, such as car doors and hummingbird wings (Olah, Mordvintsev, et al., 2017). While efficient for representation learning, polysemanticity poses significant challenges for mechanistic interpretability, making neuron-level ablations or visualisations hard to attribute to a single concept.

Let $x \in \mathbb{R}^d$ be the activation vector of a layer expressed as a linear combination of k (potentially $k \gg d$) latent feature directions collected in $F \in \mathbb{R}^{d \times k}$, i.e.

$$x = F s, \quad s \in \mathbb{R}^k, \text{ sparse.} \quad (2.3)$$

Because $k > d$, the same neuron must participate in multiple columns of F , giving rise to *polysemantic* ‘units’. Defining a linear read-out $a = W^\top x$ with $W \in \mathbb{R}^{d \times d}$, we obtain

$$a = W^\top F s =: M s. \quad (2.4)$$

A neuron i is polysemantic when the i -th row of the mixing matrix M has significant weight on multiple features (row sparsity $\|M_{i,:}\|_0 > 1$). The toy model of

(Elhage, Hume, et al., 2022) shows that packing many sparse features (k large, s sparse) into a narrow layer of width d forces an unavoidable trade-off:

$$\mathbb{E} \|a - Ms\|^2 \propto \frac{\lambda k}{d},$$

where λ is the expected sparsity of s . Methods such as Sparse AutoEncoders (SAEs) reduce the effective λ by encouraging even sparser codes, thereby converting polysemantic ‘units’ into more interpretable, monosemantic directions.

Recent approaches such as SAEs (Bricken et al., 2023; Gorton, 2024), sparse probing (Dalvi et al., 2021), and neuron-level interventions (Meng et al., 2023a), combat superposition by learning over-complete sparse dictionaries or low-dimensional subspaces that disentangle features. In particular, SAEs compress activations into sparse, over-complete features that split polysemantic neurons into monosemantic latents and have revealed interpretable features in both CNNs and transformers (Conerly et al., 2024).

2.2.3 Sparse Autoencoders

Dictionary learning (Olshausen and Field, 1996) provides a principled approach to resolving superposition. The goal is to discover an overcomplete dictionary $D \in \mathbb{R}^{d \times k}$ (where $k > d$) whose columns serve as basis vectors that can represent input data $x \in \mathbb{R}^d$ through sparse linear combinations:

$$x \approx D \cdot \alpha,$$

where $\alpha \in \mathbb{R}^k$ is a sparse coefficient vector. Unlike standard bases that require exactly d linearly independent vectors, dictionary learning trades redundancy (over-complete basis) for sparsity (each data point activates only a few basis vectors). This sparse, distributed representation disentangles features that would otherwise be superposed in the original d -dimensional space, as described by Equation 2.3.

Classical algorithms such as K-SVD (Aharon et al., 2006) and FISTA (Beck and Teboulle, 2009) solve this optimisation problem through iterative sparse coding and dictionary updates. SAEs provide a neural network approach to this same objective.

SAEs utilise neural networks to perform unsupervised discovery of the dictionary. This provides a sparse encoding of the original features of the model (which we refer to as latents in the SAE), which arise due to the enforcement of a sparsity penalty applied during the training of the SAE. SAEs aim to faithfully reconstruct the activations of a given layer in a network while keeping their own internal activations sparse. The training objective is therefore to find encoder and decoder weights that minimise a loss function combining a reconstruction term and a sparsity term, which typically takes the form:

$$L(x) = \|x - \text{Dec}(\text{Enc}(x))\|_2^2 + \lambda \|\text{Enc}(x)\|_1$$

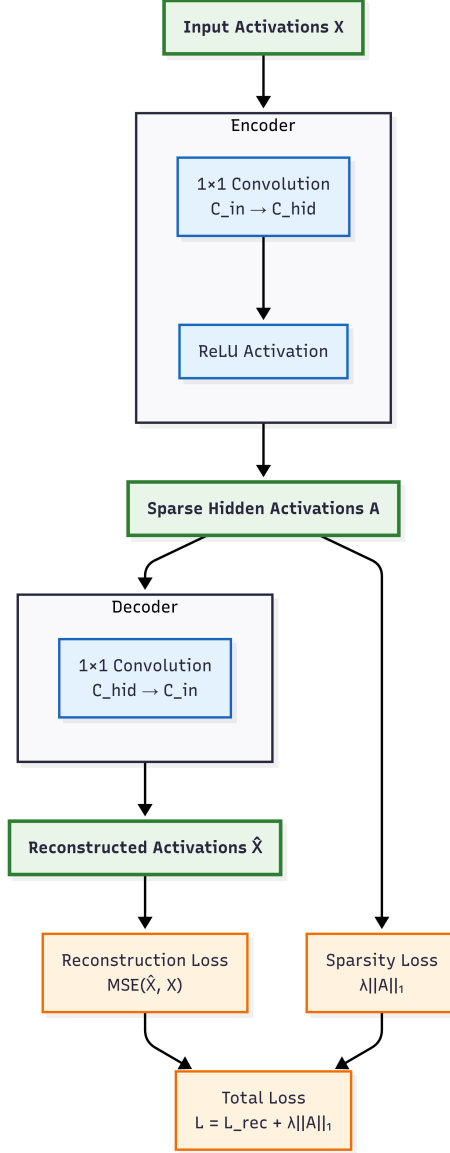


Figure 2.2: Sparse auto-encoder (SAE) architecture. An encoder applies a 1×1 convolution and ReLU to map input activations X to sparse hidden activations A ; a second 1×1 convolution reconstructs $\hat{X}$. Training minimises the reconstruction loss $\text{MSE}(X, \hat{X})$ plus an ℓ_1 sparsity penalty $\lambda \|A\|_1$. *Shapes:* $X \in \mathbb{R}^{B \times C_{\text{in}} \times H \times W}$, $A \in \mathbb{R}^{B \times C_{\text{hid}} \times H \times W}$, $\hat{X} \in \mathbb{R}^{B \times C_{\text{in}} \times H \times W}$.

Here, x represents the original activation from the base model, the L2 norm penalises poor reconstruction, and sparsity is enforced by minimising the product of the L1 norm of the SAE’s internal activations, and a coefficient λ (Bricken et al., 2023).

SAEs typically learn an overcomplete basis, where the hidden layer of the SAE is much larger than the original layer it is reproducing, typically chosen by multiplying the original hidden dimension size by some expansion factor. This high dimensionality allows the SAE to represent the input activations using a small number of highly specific, disentangled features. This addresses the problem of polysemantic ‘units’ in a model, which can make the task of accurate interpretability much more challenging (Elhage, Hume, et al., 2022).

The direct methodological precedent for applying SAEs to convolutional networks comes from recent work by (Gorton, 2024), who successfully trained SAEs on the InceptionV1 image classifier. Their approach uses a ‘1x1-conv SAE’ architecture, as illustrated in Figure 2.2. An encoder with a learnable 1×1 kernel maps the input activation tensor X to a sparse, high-dimensional hidden activation tensor A , which a symmetric 1×1 decoder then uses to reconstruct $\hat{X}$. To train this effectively on CNNs, they introduced several key innovations. First, to manage computational cost, they used *activation oversampling*, preferentially sampling training data from spatial locations with high activation magnitudes to focus on learning meaningful features rather than background noise. Second, they used highly over-complete dictionaries, with expansion factors up to 8, to provide the capacity needed to disentangle features. The results were compelling: their SAEs successfully split known polysemantic ‘double-curve’ units in InceptionV1 into distinct monosemantic features and recovered a suite of previously ‘missing’ curve detectors. These findings provide concrete evidence that dictionary-learning SAEs can resolve superposition in CNNs and motivate their application to RL agents with convolutional backbones.

2.2.4 Circuits

The circuits framework by (Olah, Cammarata, et al., 2020) provides methods to understand how neural networks combine simple features into complex representations. Their work on polysemantic ‘units’ (neurons responding to multiple distinct features) is particularly relevant when an agent must represent multiple objectives (Elhage, Hume, et al., 2022). These circuits tend to feature in most complex neural networks as implementations of algorithms that the model has learned to do work. An example of this is given by (Cammarata et al., 2020), showing how specific parts of the network learn to detect curves in images, and how these early curve features are composed together in constructing features critical to understanding circles and other curved objects in images so that they can be correctly classified.

These circuits are a critical aspect of understanding neural networks and thus MI, and identifying components that make up broader circuits in the network is essential to correctly characterising its operations.

2.2.5 Feature Visualization Techniques

Techniques such as those developed by (Zeiler and Fergus, 2013) represented the first work on interpreting the operations of deep neural networks. They employ a deconv-net, a model that maps feature activations from a deep layer back to the input pixel space by approximately reversing the network’s pooling and convolutional operations. By reconstructing the maximally activating zones in pooling layers and being chained through subsequent CNN layers the reconstruction of complex visual representations of deep layers within the network can be achieved. This is normally very challenging due to the significant compression that is performed on the deep layers in the CNN. By applying this deconvolution the features appearing deeper in the network can be expanded back up to human interpretable scales.

Early work in understanding the functioning of neural networks focused on techniques that could visually represent the features that were encoded in the weights of the neural network. In particular, (Olah, Mordvintsev, et al., 2017) developed techniques to generate maximally activating synthetic inputs that functioned as startlingly clear representations of channels and layers within the network. These visualisations revealed a clear hierarchy of abstraction, demonstrating how simple features from one layer (‘curve-circuits’) compose into more complex, semantically coherent representations (e.g., object parts) in the next. This work also demonstrated the use of gathering maximum activating samples from the training data and pairing it with particular channels and showing the similarity between these and the synthetically generated examples as a means of confirming the veracity of the visualisations.

While synthetic visualisations can be noisy, maximally activating patches, especially from within the environment, can yield interpretable results, confirming findings in (Hilton et al., 2020). Work done on exploring SAEs trained on CNNS such as (Gorton, 2024) makes effective use of visualisation to illustrate the improvement in interpretability of the SAE latents.

2.2.6 Post-hoc Explanations

Attribution and saliency methods

Post-hoc explanation methods aim to provide insights into already trained models without altering their architecture or training process. Examples include probing, SAEs, feature attribution methods (like LIME (Ribeiro et al., 2016), SHAP (Lundberg and Lee, 2017)), saliency maps (Selvaraju et al., 2017; Simonyan et al., 2014),

and activation steering (Turner et al., 2024b; Zou et al., 2025).

LIME approximates a model’s local behaviour around specific inputs using interpretable linear functions (Ribeiro et al., 2016), while SHAP assigns feature importance values based on Shapley values from cooperative game theory, calculating each feature’s average marginal contribution across all possible combinations (Lundberg and Lee, 2017). Both methods provide feature-level explanations but are computationally expensive and do not enable direct behavioural manipulation.

Saliency-based feature-attribution techniques compute the gradient of a policy output with respect to the input observation, highlighting pixels that most strongly influence the agent’s decision (Greydanus et al., 2018). Extensions such as integrated gradients (Sundararajan et al., 2017) and policy saliency maps (Atrey et al., 2020) have been applied to RL. However, these methods have significant limitations: they provide only correlational evidence of feature importance, do not enable behavioural manipulation, and critically, have been shown to fail basic sanity checks, producing visually similar outputs even when model weights are randomised, calling into question whether they truly reflect learned representations (Adebayo et al., 2020). This highlights a common pitfall in MI research, that it is very easy to convince oneself of a narrative when presented with partial data from the model, but these can widely diverge from reality.

Probing

Perhaps the most famous and enduring of post-hoc techniques is probing, particularly the linear probe (Alain and Bengio, 2018). This is simply a linear classifier trained on activation data from a given layer on target labels that are provided based on a pre-registered target of interest. The degree of accuracy that the probe can achieve represents the degree to which a given layer can be said to contain a representation of the given target feature within the layer (Alain and Bengio, 2018). Deeper probes consisting of multiple layers are also a useful tool, but these can obfuscate the degree to which the target feature is actually present due to the additional computation being performed.

Probing is a core tool in mechanistic interpretability due to the wide array of situations where it can be applied and the low cost of implementation. It also reveals a great deal of information about where in the network a certain operation is being performed such as object detection, spatial reasoning, or feature composition.

An instructive example comes from Othello-GPT (K. Li, Aspen K. Hopkins, et al., 2024), where a model trained to predict legal moves was shown to encode board state internally. Initially, 2-layer MLP probes achieved 98.7% accuracy extracting piece positions while linear probes achieved only 20%. However, (Nanda et al., 2023) showed that linear probes could match this performance when labels were

changed from ‘black/white’ to ‘mine/yours’, revealing that the MLP probes had been performing additional computation rather than simply reading off representations.

However, probes do have significant limitations as well. They provide only correlational evidence of the purpose of what the network is doing. They may show that some activation contains the relevant information but not what it is doing with it or whether it is acting upon it. There is also a risk that probes are in fact doing computational work to construct features rather than simply revealing them, if the probes have multiple layers and many neurons. For example, an MLP probe trained to detect the presence of a cat in activation space could infer the cat from isolated features in the network that would normally only be combined later in the network. The Othello-GPT example illustrates how the MLP probes were performing additional computation to adjust to the non-native representation of the board state within the activations.

Sparse Probing

While standard probing uses all activations in a layer, sparse probing is a variant that addresses polysemanticity by training linear classifiers that use only a subset of k neurons to predict target features (Gurnee, Nanda, et al., 2023). When $k = 1$, this isolates individual feature-detecting neurons, with some achieving over 75% accuracy on high-level semantic features. However, sparse probing requires significant human effort to interpret detected features, and unsupervised dictionary learning approaches such as SAEs have become the preferred method for isolating features in neural networks.

Concept Activation Vectors

Concept Activation Vectors (CAVs) bridge the gap between neural network activations and human-understandable concepts (Kim et al., 2018). To construct a CAV, one collects examples exhibiting a concept (e.g., ‘striped’) alongside counter-examples, records activations from a chosen layer, and trains a linear classifier to separate them. The normal vector to the decision boundary becomes the CAV, which can be used to test how much the network relies on that concept for its predictions, or to amplify/suppress the concept during inference. This technique laid the conceptual foundation for steering vectors (Turner et al., 2024b).

Activation Steering

Activation steering, as demonstrated in (Mini et al., 2023; Turner et al., 2024a) discussed further in Section 2.5, is another post-hoc technique. Steering typically makes use of CAVs, captured by using the difference in activations between states with and without a particular characteristic, can be used to influence a

policy model’s behaviour by acting as a superstimulus for particular environmental aspects. This is achieved through the transformation $h^l = h^l + \alpha \cdot v^l$, where the steering vector v^l is added to the original activations h^l , with α modulating the intervention strength. The steering vector itself is computed as $v^l = \mathbb{E}[h^l|\text{with target}] - \mathbb{E}[h^l|\text{without target}]$, capturing the difference in activations when the target is present versus absent.

In language models (LMs) a typical approach used to do activation steering is to use contrastive pairs (Turner et al., 2024a). In this approach one presents two different segments of text that are identical in every way except one, for example one may pass the following two inputs through an LM and then calculate the mean across the activations of the tokens and then calculate the difference in the two mean values to capture the ‘speak in French’ concept vector, which can then be used to steer the model to speak in French or English by modifying α to be positive or negative and adjusting its magnitude. This is normally done across many samples, with the mean of those pairs being used for steering.

- **English:** “The teacher explained the complex problem using simple examples.”
- **French:** “L’enseignante a expliqué le problème complexe en utilisant des exemples simples.”

Activation steering is an excellent tool for interpretability due to the ease and speed with which it can be used, and because it allows researchers to immediately observe how a network responds to a specific concept being made more or less prominent in the forward pass of the network. They also allow researchers to determine whether a concept exists and is easily representable in given layer of a network. Given that you can extract concept from single layers, or multiple layers simultaneously, one can test whether the concept is meaningful within a certain part of a network by the degree of change in output observed by its application there.

Activation Patching

Activation patching, also known as causal tracing, is a related but distinct technique from activation steering. While steering adds a direction vector to shift activations ($h' = h + \alpha v$), patching replaces activations from one forward pass with those from another to identify which components are causally responsible for a behaviour (Meng et al., 2023b).

The ROME (Rank-One Model Editing) framework (Meng et al., 2023b) introduced systematic causal tracing to localise where factual associations are stored in language models. By corrupting input embeddings and then selectively restoring activations at specific layers during the forward pass, they identified that mid-layer

MLP modules in GPT models are critical sites for factual recall. This ‘corrupt-then-restore’ methodology provides a template for identifying causal bottlenecks in neural networks.

In RL contexts, (Mini et al., 2023) demonstrated that clamping a single activation to a fixed value at a specific spatial location within a convolutional channel could reliably redirect a maze-solving agent’s navigation. By setting one activation out of tens of thousands to a target value, the agent would navigate toward the corresponding maze location as if a goal were present there. This spatial patching approach exploits the correspondence between activation positions in convolutional layers and locations in the environment, providing direct causal evidence that specific channels encode navigational targets.

Many of the interpretability techniques discussed above were developed for supervised settings with independent and identically distributed (i.i.d.) data. Reinforcement learning violates these assumptions: observations are generated by an agent acting in a dynamic environment, and credit must be assigned across time. As a result, many tools require adaptation to work within this paradigm.

2.2.7 Causal Methods in Interpretability

Many of the techniques discussed in the methodology, such as activation steering (Section 2.2.6), can serve dual roles as both post-hoc explanation tools and causal testing methods when applied systematically. This section focuses on additional causal techniques and frameworks for rigorous causal claims in interpretability research. These techniques actively modify the operations of the network, testing whether specific components are genuinely responsible for behaviours of interest (Pearl, 2009).

Modern systematic ablation has evolved through several generations. Zero ablation, the simplest approach, sets components to zero but can lead to out-of-distribution behaviours since networks rarely encounter zero values during training. Mean ablation, which replaces components with their average values across a sample dataset, better preserves activation statistics but may incompletely remove information if the mean itself carries useful information. Recent work (M. Li and Janson, 2024) introduces optimal ablation constants that are computationally determined to minimise the expected loss on the task, reducing the amount of distributional shift in the final outcome.

For agents with spatial convolutional representations, targeted synthetic interventions can directly test the functions of parts of the network by intervening on their values directly. (Mini et al., 2023) demonstrated that clamping a single activation in a channel that typically identified the location of the cheese (channel 55) to a large positive value at a specific location in the maze could steer the agent to that location, as if that was the final goal of the maze. This provided direct causal

evidence that this channel had an unusual ability to steer the direction of the agent.

These causal techniques provide the foundation for the experimental methodology employed in this thesis, moving beyond observation to establish genuine causal mechanisms underlying goal representation in RL agents.

2.3 Interpretability in Reinforcement Learning

Interpretability in RL presents unique challenges. The agent’s interaction with an environment means that specific outputs of the model may be semantically distant from the contents of a given activation. Furthermore, we can only infer the meaning of an activation within the model; we cannot directly glean it as we could a word extracted from the embedding space of a language model.

Recent research shows that such models naturally learn geographic mappings of the world related to locations that can be retrieved directly from the activations of the network, effectively building simplified world models (Gurnee and Tegmark, 2023). Similarly, analyses of agents trained on complex games like chess demonstrate that these systems acquire human-understandable strategic concepts, such as piece values or tactical motifs, purely from self-play (McGrath et al., 2022). These findings suggest that neural networks learn richer internal representations than previously thought, lending support to the idea that goal representations can be directly extracted and understood. Figure 2.1 illustrates how even a seemingly simple maze environment can embed multiple, dynamically changing goals whose representations the agent must learn to use effectively.

Work by (Betley, 2023) found a specific circuit in the convolutional layers of the Impala model that activated strongly when the direction to go was ‘up’ and didn’t activate when the goal was in a similar position but the agent would need to go in a different direction. This indicates that much of the ‘planning’ was already performed in the CNN layers, suggesting it should be possible to train probes that can detect the exact direction the agent needs to move towards.

Work exploring the Procgen Coinrun environment builds on saliency based methods (Section 2.2.6) to calculate an attribution within the model weights themselves. By measuring the contribution of a given convolutional feature map to the value function, it becomes possible to determine whether a particular feature is positively or negatively associated with task success. This is computed by multiplying the gradient of the value function with respect to each activation by that activation’s value. This advances previous saliency mapping methods by linking the gradients to specific features within the model, rather than pixels in the input (Hilton et al., 2020).

This is achieved by applying non-negative matrix factorisation (NMF) to the resulting attribution tensor (the element-wise product of activations and gradients across all spatial locations and channels) to cluster correlated locations into object-level ‘features’ such as coins, enemies, and obstacles. By using this dimensionality reduction they are able to produce clear, interpretable object detectors.

2.4 AI Safety

AI safety research aims to ensure that AI systems reliably produce intended outcomes. This section reviews the key failure modes relevant to RL agents and how mechanistic interpretability can help address them.

2.4.1 Motivation and scope

AI Safety can primarily be framed as a question of how to reliably obtain the outcomes that we intended from the deployment of AI and ways that this could go wrong, both on a technical and social level. Because AI systems are complex and powerful, this is a challenging and moving target. In practice this means that the work of ensuring good outcomes from AI requires a multi-pronged approach and a high degree of involvement in order to prevent failure and effectively intervene when models fail in unexpected ways (Hendrycks et al., 2023).

One useful lens for analysing risks from AI (and risk in general) involves considering the multiplicative formula from (Hendrycks et al., 2023):

$$\text{risk} = \text{hazard} \times \text{exposure} \times \text{vulnerability}$$

This formula helps us consider different interventions in terms of which of these terms we aim to reduce, and to concretely estimate the impact of a given intervention on the downstream risks.

Hazard, in the above equation, is the amount of potential damage that something could do under the wrong circumstances and will continue to grow as AI systems become capable of greater levels of harm due to increasing capabilities. Exposure refers to the number of people that would potentially be harmed in the case of failure. Vulnerability represents a group or individual’s degree of susceptibility to the particular hazard. An example would be an underwater volcano, which by itself presents a significant hazard but because no one is exposed presents no risk. However, if one builds an undersea internet cable connecting two continents beside it then the exposure and vulnerability are significantly increased. By coating the cable in a protective casing, this reduces vulnerability, while exposure stays the same. AI safety as a field can be characterised as an attempt to reduce one or more of these metrics with respect to powerful AI systems and humanity.

2.4.2 A short history

The field of AI safety began to develop into a more recognisable version of what it is today in the late 2000s, with papers such as (Yudkowsky and Yudkowsky, 2008) laying the groundwork in describing key concerns, failure modes and impacts of powerful AI systems. This paper argued that AI deserved a similar level of concern as other more commonly acknowledged existential risks such as nuclear war or pandemics. It also laid out key arguments around why AI could be uniquely dangerous in a ‘hard takeoff’ scenario where it is able to rapidly improve its own capabilities and alter safeguards or goals that its developers had trained it to have, degrade those goals and have them shift over during recursive self-improvement. It also discussed the dangers of anthropomorphising AI systems and expecting them to behave as we might. This is most clearly argued for in the idea of the orthogonality thesis, which claims that there is no reason why intelligence and good behaviour would have an inherent relationship to each other, and that whatever goals we placed into the system could be pursued with arbitrarily high intelligence, regardless of the nature of the goal. (Omohundro, 2008) argues further that for any given task there will be an inevitable drive to pursue instrumental goals that will aid in the completion of the original goal, and that these instrumental goals are likely to lead them into conflict with humans.

First published in 2010, (Bostrom and Yudkowsky, 2014) developed the early ideas into a clearer taxonomy of near-term and long term risks, and argued that the ideas required serious and immediate attention despite the speculative subject matter. This work was published in the ‘Cambridge Handbook of Artificial Intelligence’, which helped to present the discussion to a wider audience. The final step to broad mainstream interest came with the release of *Superintelligence* (Bostrom and Bostrom, 2014).

More concrete technical research began to develop following the release of (Amodei et al., 2016), which identifies concrete problems in AI safety, which are discussed more in the next section.

2.4.3 RL Safety

RL suffers from several well-documented failure modes: specification gaming, where agents exploit loopholes in the reward signal; unsafe exploration, where agents take dangerous or irreversible actions while learning; distributional shift, where agents fail when deployed outside their training distribution; and goal mis-generalisation, where agents pursue the wrong objective at deployment (Amodei et al., 2016; García and Fernández, 2015; Krakovna, 2020; Langosco et al., 2022).

These pathologies often surface in surprising ways. A well-known example of specification gaming is the CoastRunners agent that learned to drive in circles collecting speed boosts rather than racing, because this strategy yielded higher

reward (Krakovna, n.d.).

Even if exploration is safe and training rewards are faithful, agents can fail under distribution shift. Policies that perform well in simulation or under limited training conditions often generalise poorly when the environment changes (Amodei et al., 2016). Procgen itself has been introduced as a benchmark to quantify such generalisation gaps (Cobbe, Klimov, et al., 2019), which makes it a natural testbed for the present thesis.

Relatedly, goal misgeneralisation describes policies that appear successful during training but pursue the wrong goal when deployed outside of their training distribution (Koch et al., 2021). This problem is particular to RL agents, and is challenging to mitigate because it may be unclear ahead of time what objectives the model may have internalised, and thus can lead to surprising failures during deployment.

This gap between performance on the training distribution and robust, safe behaviour has motivated recent work arguing that mechanistic interpretability is a critical component of ensuring RL safety (Bereska and Gavves, 2024; Hubinger et al., 2019; Olah, Satyanarayan, et al., 2018; Shihab et al., 2025). The premise of this research is that by directly inspecting the learned representations and algorithms inside a policy network, it may be possible to move from post-hoc observation of failures to a predictive understanding of their causes.

2.5 Understanding and Controlling a Maze Solving Policy Network

(Mini et al., 2023) demonstrated the ability to manipulate an agent’s navigation in a simple maze environment with an intervention to a single channel in the network. This work provides the primary motivation for this thesis, with the intent of extending these techniques to more complex multi-objective settings. Their work specifically used an agent trained in a modified environment where the cheese only ever appeared in the top right, then deployed this agent into settings where the cheese could spawn at any location. This created a tension between the agent’s learned tendency to navigate to the top right and its drive to approach the cheese.

They go on to show that the agent’s decision making about whether it travelled to the top right or towards the cheese alone was determined by both the Euclidean distance between the agent and the cheese and the path distance between the agent and the cheese. This means that the agent had multiple separate mechanisms driving its decisions rather than being a single unified policy. They did this by showing through regression analysis that if the agent was an equal path distance from two instances of cheese, it would tend to approach the cheese that was also

closer in Euclidean distance.

This shows a surprising contradiction in the agent because Euclidean distance should not meaningfully affect a maze-solving policy (the agent cannot walk through walls), suggesting the agent is not relying on a single reward-maximising policy but is instead operating under multiple competing goals.

They further demonstrate that neural networks can develop interpretable mechanisms to track goals. The authors identified specific channels that tracked the location of objectives in a maze environment and showed that these could be manipulated to control the agent’s behaviour. They found that steering vectors, the difference in activations between otherwise identical states with and without an entity, could be used to influence the model’s behaviour in predictable ways. This work serves as the foundation for our research, though we extend it to a multi-objective setting.

Recent concurrent work by (Trim and Grayston, 2024) extends the work done by (Mini et al., 2023) to analyse goal misgeneralisation. Their layer-by-layer analysis reveals a hierarchy of representations: early convolutional layers detect basic features such as walls, paths, and the cheese goal; middle layers develop more abstract concepts including patterns that propagate outward from objects (similar to a flood fill algorithm); while deeper layers create highly abstracted representations where identifying the goal becomes challenging. They further confirm the positional bias exists regardless of the final goal’s position within the maze, indicating a deeply learned tendency towards this direction rather than a perceptual error.

2.5.1 Discovering and Controlling Goal Representations

(Mini et al., 2023) successfully isolated the network’s representation of the ‘cheese’ goal and demonstrated causal control over the agent’s behaviour. First, they defined a ‘cheese vector’ by taking the difference in mean activations at a specific layer between trajectories where cheese was present and those where it was not:

$$v_{\text{cheese}}^l = \mathbb{E}_{s \sim \text{traj}_{\text{cheese}}}[\text{act}^l(s)] - \mathbb{E}_{s \sim \text{traj}_{\text{no-cheese}}}[\text{act}^l(s)]$$

Adding this vector back into the network’s forward pass, scaled by a coefficient α , allowed them to controllably amplify or suppress the agent’s cheese-seeking behaviour.

Their most significant contribution was demonstrating this control causally through *activation patching*. They intervened on the final convolutional layer of the policy network by copying the activation pattern from a small spatial region of a channel corresponding to a real cheese location and ‘pasting’ it into the same channel at a different, empty location. By doing this they could reliably steer the agent to the empty location as if the cheese were there.

This effect was highly localised. They identified that out of 16 channels in the layer, just 11 were responsible for this behaviour. Ablating these "cheese channels" destroyed the agent's ability to find the cheese, while leaving other navigational capabilities intact. This proves that goal representations can be sufficiently modular and concentrated to allow for targeted manipulation, motivating the use of more advanced techniques like SAEs to further dissect these representations in a more complex, multi-objective setting.

2.6 Gaps in the Literature

Several areas in RL interpretability remain underexplored, among others:

- How RL agents track and switch between multiple sequential objectives
- Whether SAEs can disentangle representations in RL agent CNNs as effectively as in supervised classifiers
- Whether activation magnitude (not just activation presence) encodes goal identity
- How interpretability findings generalise from simple maze agents to more complex architectures

This thesis addresses the first three gaps directly, investigating goal representation in a multi-objective maze-solving agent.

2.7 Summary

Prior work has established foundational techniques for understanding neural network internals, from activation visualisation and steering vectors to SAEs for resolving polysemanticity (Bricken et al., 2023; Elhage, Hume, et al., 2022; Olah, Cammarata, et al., 2020). Work on maze-solving agents (Hilton et al., 2020; Mini et al., 2023) has demonstrated that goal representations can be isolated and manipulated, providing a template for mechanistic analysis of RL agents.

This thesis extends these techniques to multi-objective settings, investigating how agents track and switch between sequential goals. We apply SAEs to RL agent CNNs, systematically analyse channel roles through ablation and intervention, and explore whether activation magnitude encodes goal identity. The following chapters detail our methodology and findings.

Chapter 3

Methodology

3.1 Environment and Model

3.1.1 The Procgen Heist Environment

The Procgen Heist environment, from the suite of procedurally generated environments by (Cobbe, Hesse, et al., 2020), was chosen for this research. The agent’s objective is to navigate a maze to collect a gem, which may require first collecting up to three keys (in the fixed order of blue, green, red) to open corresponding locks. The environment’s procedural generation varies the maze layout and the number of key-lock pairs present in each episode, from zero to three.

This environment provides a sparse reward signal, with a reward of +10 for reaching the gem and 0 otherwise. This design, coupled with the variable difficulty, establishes a natural curriculum that allows agents to succeed with random exploration in simpler levels before developing a more sophisticated policy.

The choice of a Procgen environment is motivated by the need to learn generalisable representations. The procedural nature of the mazes discourages the agent from memorizing simple heuristics and instead promotes the development of robust internal models of entities and game mechanics. Furthermore, Heist’s multi-objective nature is expected to produce complex internal behaviours, such as general-purpose navigation circuits that can be dynamically retargeted based on the current goal.

For this investigation, the ‘easy’ distribution of the environment was used, which provides the full 64x64 observation to the agent, making the environment a fully-observable Markov Decision Process (MDP). While memory-based, partially-observable variants exist, they were not used, as full observability simplifies the interpretability analysis by eliminating the need to account for memory mechanisms. The

distribution of entities is noteworthy: since levels are sampled uniformly from all configurations, the gem and the blue key/lock appear more frequently than the green and red pairs.

3.1.2 Model Architecture

We train a reduced version of the Impala model that uses 5 convolutional layers instead of 15 in the original; this is a smaller architecture, not an attempt to compress the same computational capacity into fewer layers. This architecture choice reflects established principles for interpretability research: following the precedent set by (Hilton et al., 2020), simpler architectures often yield more interpretable representations without significant performance loss. We include the full model architecture in Appendix A.

3.1.3 Model Training

The policy models were trained using Proximal Policy Optimization (PPO) with an open-source implementation based on PFRL, specifically designed for Procgen environments. We also used Envpool to accelerate training while keeping the dynamics unchanged (Weng et al., 2022). Training used a learning rate of $5e-4$ and was conducted until convergence, defined as no statistical improvement for 10,000 update steps. The primary model was trained for approximately 800 million environment steps (60,000 update steps). Training dynamics were significantly impacted by changes to batch size, often completely collapsing training performance, but over 5 runs with separate seeds we replicated similar model dynamics.

To explore the effects of training dynamics on the final learned representations, experiments were run with different training configurations. These variations primarily involved altering the batch geometry through four main hyperparameters: the number of parallel environments, the number of steps per rollout, the mini-batch size, and the number of epochs.

In the primary configuration reported in this thesis, we used a setup of ($N_{\text{env}} = 64$, $N_{\text{step}} = 256$, `batch_size` = 8, $n_{\text{epoch}} = 3$). In this geometry, the rollout buffer contains $N_{\text{env}}N_{\text{step}} = 16\,384$ transitions. This is reshaped into 8 mini-batches, each containing $B_{\text{mb}} = 2\,048$ samples. The optimiser performs $8 \times 3 = 24$ gradient steps per PPO update, processing a total of $16\,384 \times 3 = 49\,152$ sample-updates per cycle. This configuration was chosen to balance stable learning from large batches with reasonable training time per update.

We ran 5 sets of training runs for this geometry in order to confirm the repeatability of our results, and ran further training runs with other geometries to observe the extent to which this altered the final structure of the model weights. Namely ($N_{\text{env}} = 128$, $N_{\text{step}} = 128$, `batch_size` = 8), ($N_{\text{env}} = 256$, $N_{\text{step}} = 256$,

`batch_size = 8`), and ($N_{\text{env}} = 256$, $N_{\text{step}} = 128$, `batch_size = 16`). These variants were included to probe the sensitivity of the learned representations to rollout length and batch composition. A detailed comparison of the models trained under these settings is provided in Chapter 4.

3.2 Interpretability Techniques Employed

3.2.1 Activation Visualization

We visualised convolutional layer activations by applying a colour spectrum according to activation strength at each spatial location, following (Mini et al., 2023). This provided a direct view of which spatial regions triggered strong positive or negative responses in different channels. Figure 3.1 shows this technique applied to multiple layers.

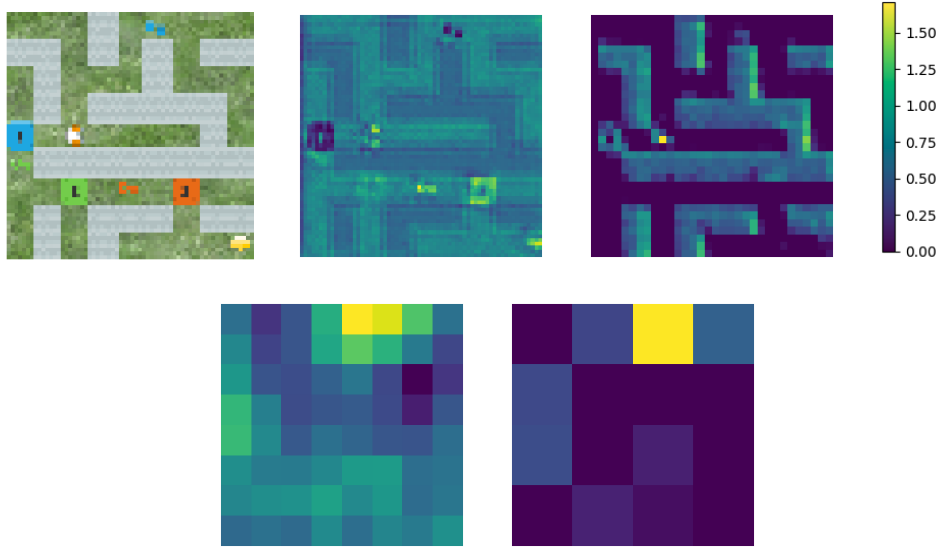


Figure 3.1: Visualisation of the maze and neural network activations in different layers. Top row (left to right): observation input, conv1a channel 10, conv2a channel 10, and activation colourbar. Bottom row: conv3a channel 29 and conv4a channel 29. In the top row filters we observe a minimal activation around the regions of the blue key and lock and in the bottom row we see maximal activations around the blue key. This is a replication of the approach used by (Mini et al., 2023) in order to get an initial sense of the clarity with which entities could be seen in activations.

3.2.2 Comparative Activation Analysis

Parallel rollout analysis

To compare activations during entity targeting, we created a simple toy maze to test basic capabilities in the model, specifically its ability to turn a corner and travel in a straight line towards a target. The maze took the shape of a T junction, with the agent started at the middle point, and with a blue key placed in the bottom right. A visual of the maze is provided in Figure 3.2. A rollout of the policy was then performed, but for each step of the rollout, the environmental state was modified to produce an observation with the gem, blue key, green key, and red key. By doing this at every step, 4 parallel rollouts with different entities were produced which allows for a perfectly controlled experiment from which we could observe the variations in activation patterns based only on the nature of the target entity.

The purpose of this experiment was to determine to what extent the network relies on the activation range of values as part of determining which stage of the maze it is in and which entities to target. This addresses RQs 1, 2 and 4 by isolating how the network encodes different goal entities and determining whether the model uses entity-specific channels or a general targeting mechanism.

With these rollouts we tracked the raw activation magnitudes across all channels in layers conv3a and conv4a. This design isolates the effect of entity type on the model’s internal representations while holding agent position, maze geometry, and navigational requirements constant across all four parallel trajectories.

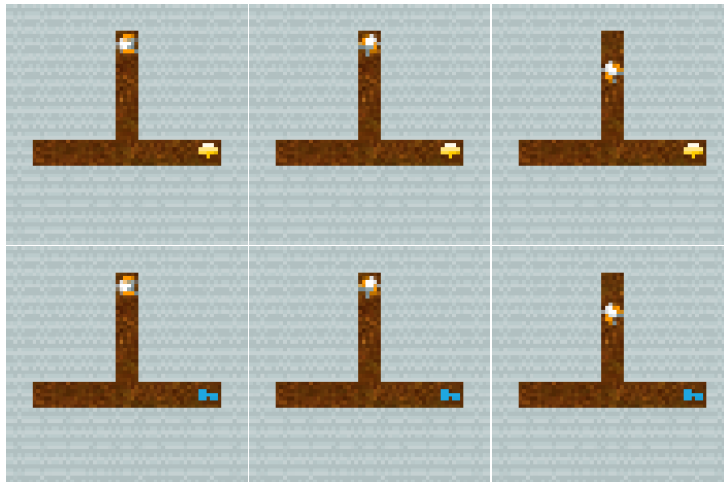


Figure 3.2: This illustrates the parallel rollout of the agent, with each step taken by the agent remaining exactly the same while we create an observation with each of the blue key, green key, red key, and gem.

Analysing activation changes over a rollout

To further investigate RQs 1 and 2, we examine how activation patterns evolve as entities are collected, revealing how goal encoding changes dynamically during task progression. The previous experiment showed how activations vary when entities are changed but all other variables are held constant. However, it is limited in that it only contains a single entity at a time, while normally the environment will have multiple entities present. To observe whether activation ranges were meaningfully related to which entities were present, we captured mean activations for given channels in a layer over the course of an episode in an unmodified maze. This allowed us to measure whether clear patterns emerged between the activations of different channels as different entities were collected.



Figure 3.3: This illustrates the 4 directional cross maze used for analysing which entity would be collected first when different activation offsets were applied. Within a given maze, the positions of the entities within a given arm of the maze would be randomised so as to produce greater permutations to allow meaningful statistical testing.

Applying offsets to channels for entity steering

The following experiment aims to address RQ2 by testing whether activation magnitude alone can alter entity targeting, and contributes to answering RQ5 by measuring the ability to affect steering across different entities. To do this we swept across a parameter α , which represented a constant offset that we would add to all channels in a given layer simultaneously. We would then perform a rollout in a

custom four armed maze environment, which can be seen in Figure 3.3. This maze contains the keys and the gem but no locks, and allows for placement of entities at multiple random points within the arms to allow for statistical variance between runs. We used this environment because it provides some navigational challenges, but also makes it possible for the agent to go towards any of the entities, meaning we can observe its preferences. We also remove any confounders that may arise from specific maze structures.

To determine the extent to which uniform activation offsets affect target selection, we performed a comprehensive sweep by simply adding a constant offset α to all channels in a given convolutional layer.

To describe this formally: for a given maze seed s , channel c in layer l , and at each timestep t , we modify the channel activation $h_t^{c,l}(s)$ as follows:

$$\tilde{h}_t^{c,l} = h_t^{c,l}(s) + \alpha \quad (3.1)$$

Here, α is the flat scalar offset applied. We performed parameter sweeps over α in the range $[-15, +15]$ in increments of 0.2, with $n = 100$ offset values per increment for conv4a. Some layers exhibited changes over a much narrower activation range such as conv3a and conv2a, and for these we performed sweeps in the range $[-3, +3]$ with increments of 0.03.

3.2.3 Sparse Autoencoder (SAE) Training

We trained SAEs on the CNN layers of the compressed IMPALA model to test whether apparent polysemanticity could be decomposed into cleaner features (RQ3). Our methodology builds upon (Gorton, 2024), using their 1×1 -conv recipe with an L1 warm-up schedule.

Training ran for 6M steps, during which variance explained exceeded 99% and policy reward recovered to within 99% of baseline. A compressed variant (conv4a reduced to 20 channels) was also trained to test robustness under capacity constraints. Table 3.1 lists expansion factors and L1 coefficients for all layers. We trained 5 SAEs per layer to verify robustness.

Hyperparameter selection followed conventions established in prior work on CNN SAEs (Gorton, 2024), who used expansion factors of $2\times$ for early convolutional layers and $4\times$ for middle layers when applying SAEs to InceptionV1. We adopt similar values: a $2\times$ expansion for conv1a (the earliest layer, which learns low-level features requiring less disentanglement) and $4\times$ for deeper layers. This modest expansion avoids excessive feature splitting while still providing sufficient capacity to decompose polysemantic neurons. L1 coefficients were tuned per-layer to balance reconstruction fidelity against sparsity: early layers used smaller coefficients ($1e-4$)

to preserve fine-grained spatial information, while deeper layers tolerated larger coefficients ($5e-3$) as their representations are more abstract and naturally sparser.

Table 3.1: Expansion factors and L1 Coefficients for the different SAEs

Layer	Expansion Factor	L1 Coefficient
conv1a	2	$1e-4$
conv2a	4	$5e-4$
conv2b	4	$5e-4$
conv3a	4	$5e-4$
conv4a	4	$5e-3$

3.2.4 Validating results with SAEs

To distinguish between unstructured polysemanticity and structured multi-entity encoding, we applied the same patching and ablation techniques to SAE latents as to base model channels. If the base model’s multi-entity responses resulted from polysemanticity, SAEs should decompose them into cleaner per-entity features. If instead the multi-entity encoding represented a fundamental strategy, SAE latents should preserve similar patterns.

We searched for individual SAE latents that generalised across entity classes (e.g., a single latent redirecting the agent towards any key) versus latents dedicated to specific entities. Latents encoding multiple entity types would indicate the model learns general goal-oriented features rather than exhibiting simple polysemanticity.

3.2.5 Intervention experiments

A critical goal for this work was to uncover causal mechanisms behind the model’s behaviour and to understand how the network specifies which entity to target in a given state of the maze. These causal intervention experiments address RQ2 by identifying which network components determine entity targeting, RQ4 by testing whether specific channels control specific entities, and RQ5 by measuring differential steering effectiveness across entities. While the goals in the environment would always be sequential, there were still many cases where the model would fail if it tried to target the wrong entity, and so this was a crucial mechanism. Early observational experiments also demonstrated that the preference hierarchy for targeting the blue key, the green key, the red key, and then the gem were strongly encoded in the network.

To demonstrate a true causal mechanism we approached the problem by trying to find parts of the network that seemed to play a key role in deciding model direction. For this we used an activation patching approach. The experimental

setup is as follows: We would place a specific entity into the environment and then intervene on a single channel in a single layer by applying a zero mask to the channel and then setting a single neuron of the channel to a specified value. In this experiment, we would do a sweep over a range of values between 0 and the maximum values that the channel would reach during typical functioning as determined by sampling from the environment during operation. By running this sweep, we can get a sense of what the different channels do at different intervention strengths. We kept the length of the episode to 20 steps to ensure that the agent would either be able to reach the target zone, or the actual entity, but not both.

We use a ‘Q’ shaped maze with the design visible in Figure 3.4 because it was challenging enough that the agent would need to retain its navigational abilities to reach either target, while providing a clear decision point where the agent would need to choose between pursuing the original entity or our artificial target zone.

We test a single entity at a time in this way meaning that steering that is effective with a particular value on a given channel may not be effective for redirecting the agent in the presence of another entity.

We scaled up this approach by running a static intervention across every channel in the SAE, and in the base layer of the model. For each entity-channel combination, we performed $N_{\text{trials}} = 500$ trials. In trial i , the activation values in a target region of the channel were set to a value v_i , calculated as:

$$v_i = i \cdot \frac{V_{\text{max}}}{N_{\text{trials}}}, \quad \text{for } i \in \{1, \dots, N_{\text{trials}}\}$$

where V_{max} is the maximum activation observed for the layer. This allowed us to measure the success rate of steering as a function of activation strength and to identify whether channels influence navigation only within specific value ranges.

We test this approach across all layers in the environment. Because conv4a yielded the most promising layer due to its naturally fairly sparse representations we performed additional experiments across 28 model checkpoints over a single base model training run in increments of 3000 update steps from 3000 to 60000. We observe model convergence to maximum rewards at around 20000 steps, and thus many of the later checkpoints are variations of different equally high scoring policies.

To establish a baseline for normal activation ranges, we calculated the 99.9th percentile of activations for each channel over 50 rollouts in unmodified mazes. This threshold is used in later analysis to identify whether successful interventions operate within or outside the network’s typical activation distribution.

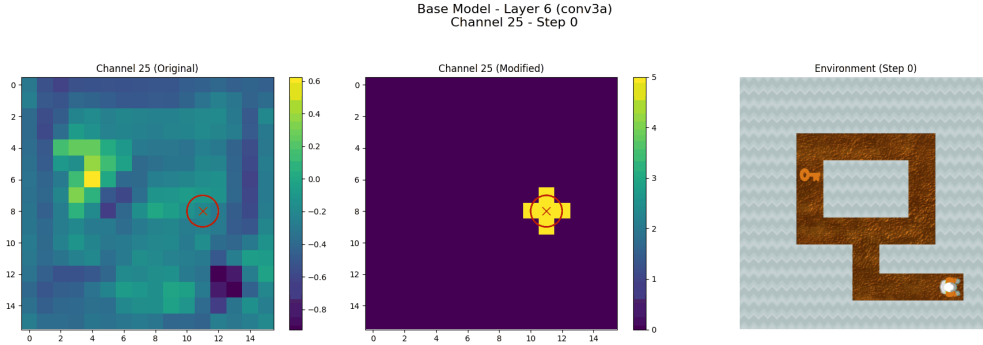


Figure 3.4: Example of the patching experiment. We test channels to determine the extent to which clamping a single activation can redirect the agent to another location given the presence of a single entity in the environment. We consider an intervention a success if the agent enters the modified region. The maze is a 9 by 9 grid, while the conv4a channels have dimensions of 8 by 8, leading to a slightly imperfect mapping.

3.2.6 Ablation Studies

To investigate the functional roles of different channels, we conducted extensive ablation experiments following the causal methods discussed in Section 2.2.7. These included: (1) testing individual channels in isolation by masking all other channels to zero ($n=400$ rollouts per channel), (2) ablating single channels to identify those critical to overall network function, and (3) targeted ablations of channel groups based on their roles in steering and navigation. We tested these ablations in controlled maze environments, including a fork maze requiring explicit path choices and an open maze with entities in corners. Full methodological details and results are provided in Appendix G. As discussed in Section 4.6, these experiments yielded complex results that did not clearly illuminate the network’s core mechanisms, though they did confirm that navigation is redundantly encoded across channels.

3.3 Probing for Abstractions

To determine at which layer in the network different concepts were processed, we performed two different probing experiments: one targeting which entity was the current correct target of the agent, and one targeting which direction the agent needed to move to reach its goal. These probing experiments address RQ1 by localising where goal information is encoded, and RQ6 by revealing which layers specialise in processing target identity versus navigational planning.

The key question we wanted to answer was at which layer the network most ac-

tively processes information about the current target. Probes are an imperfect mechanism for this because they provide only correlational evidence; information about which entity is the current target may be present at a layer without being actively used in downstream processing. However, probes do reveal whether sufficient features exist to distinguish between target classes, which can inform important intuitions about the network.

3.3.1 Next Entity Probes

This probing approach addresses RQs 1 and 2 by identifying which layers encode the next target entity, and RQ6 by comparing representational clarity across network depth. To investigate where the network encodes information about which entity to target next, we trained multi-class linear probes to predict the next target entity from layer activations.

For determining the next entity, we gathered observations from many unmodified maze rollouts, capturing activations every 7 steps to ensure sufficient temporal distance between samples while remaining computationally tractable. For each of seven entity types (blue key, green key, red key, gem, blue lock, green lock, red lock), we gathered 2000 observations where that entity was the agent’s next target, yielding 14,000 total samples balanced across entity types. This sample size was chosen to provide ample training data for robust classifier performance.

We trained 7-way classifiers to predict which specific entity was the next target. We used a 90/10 train-test split.

We trained two types of probes to distinguish between localised and distributed representations:

- **Per-channel probes:** For each channel in layers conv1a-conv4a, we flattened the spatial activations and trained linear classifiers on the flattened activations. This tests whether individual channels encode target identity.
- **Whole-layer probes:** For full layers (conv1a, conv2a, conv3a, conv4a, fc1, fc2, fc3), we flattened all activations across all channels/neurons and trained a linear classifier for each layer. This measures the total information available at each layer, regardless of how it is distributed across channels.

All probes were linear classifiers trained with the Adam optimiser (learning rate 0.01) for 200 epochs, which we found sufficient for convergence across all probe configurations.

3.3.2 Direction probes

To establish which parts of the network were more involved with navigation, we trained multi-class linear probes to predict which direction the agent would need

to move to reach the next target entity. This is a complex task because the required movement direction often differs substantially from the entity’s position relative to the agent due to maze walls. By probing for directional information separately from entity identity, this experiment addresses RQ6 by testing whether navigation processing is functionally separated from goal representation in the network architecture.

Building the training dataset involved using Dijkstra’s algorithm to calculate the most efficient route to the next entity and determine the optimal first move. For each of the four cardinal directions we gathered 2000 observations and extracted model activations, providing ample training data as with the entity probes. We excluded examples where the agent was directly adjacent to the target entity (making direction trivial) or where the agent was positioned ambiguously between grid squares (making the correct direction unclear).

We trained 4-way classifiers with the same architecture and training procedure as the entity probes.

3.4 Summary

The methodologies above address our research questions through complementary approaches. Parallel rollout analysis and activation tracking reveal how the network encodes different goal states (RQ1, RQ2). SAE training and validation test whether multi-entity responses reflect polysemanticity or structured encoding (RQ3, RQ4). Intervention experiments establish causal relationships between activations and behaviour (RQ2, RQ5). Probing experiments localise goal and navigation information across layers (RQ1, RQ6). Together, these methods provide both correlational and causal evidence for understanding goal representation in the network.

Chapter 4

Results

All results presented are from checkpoint 35001 (35k update steps), past convergence which occurred around 30k, unless otherwise noted.

4.1 Parallel Rollout Activation Analysis

Parallel rollouts reveal distinct activation trajectories for each entity type (Figure 3.2). The same channels are active across all entities, but at systematically different levels depending on which entity is being targeted. This distinguishes between two possible encoding strategies: entity-specific channels that uniquely fire for particular entities, versus shared channels where activation levels shift systematically with game state.

Clean parallel trajectories with significant gaps between entity types indicate strong value encoding for entity targeting. Channels with the lowest inter-entity offset show similar parallelism, though certain entities (e.g., red key and blue key) sometimes overlap, indicating less differentiation in those channels.

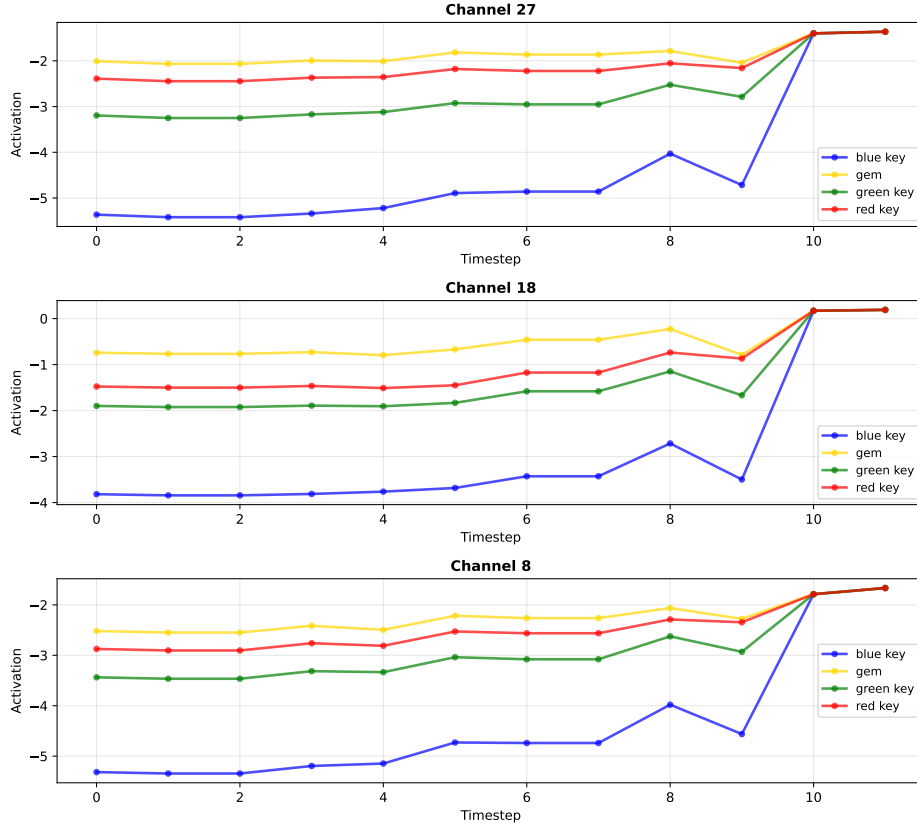


Figure 4.1: Mean activation trajectories for channels with highest inter-entity variance in layer conv4a at checkpoint 35k. Each subplot shows a single channel, with coloured lines representing the four parallel rollouts (blue key, green key, red key, gem). The x-axis shows timesteps within the rollout; the y-axis shows mean channel activation. Parallel, vertically-separated trajectories indicate that the same channels are active across entities but at systematically different activation levels depending on which entity is being targeted.

We observe extremely high correlation between activation patterns in both conv3a and conv4a. Conv3a shows an average correlation of 0.956 (range: 0.837-0.995). Conv4a has a comparable average (0.931) but contains a subset of channels with substantially lower correlations (range: 0.401-0.995). These low-correlation channels exhibit activation patterns that vary over time in ways that differ between entities, rather than maintaining a constant offset throughout the trajectory, suggesting they play an unusual role in entity targeting. The full results are shown in Figure 4.2.

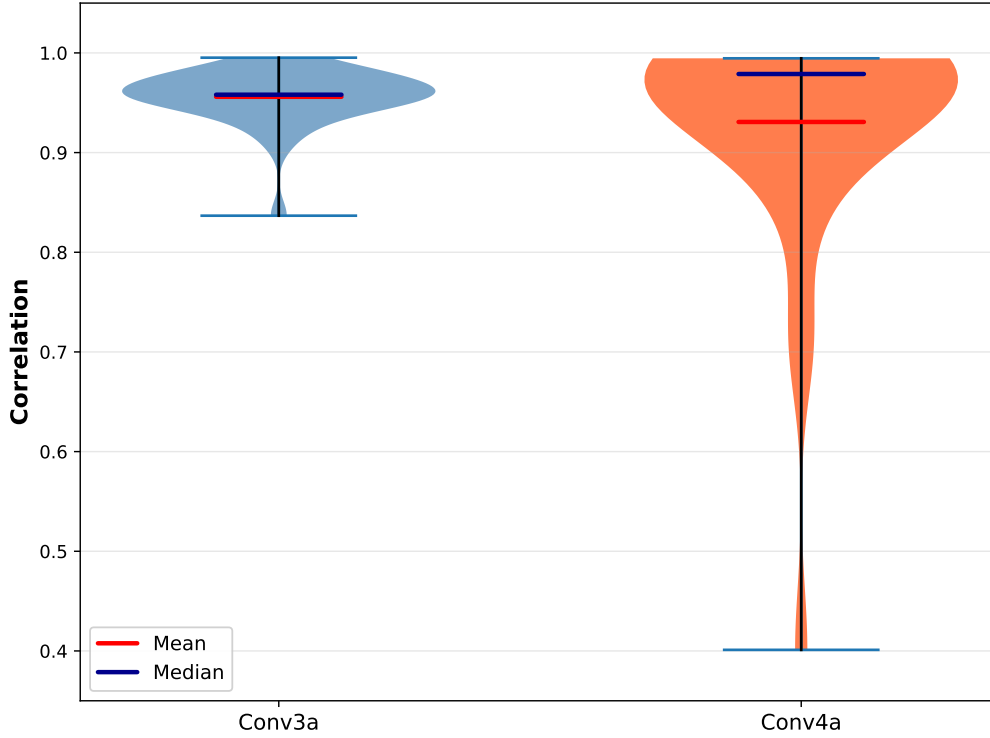


Figure 4.2: Violin plots comparing the distribution of inter-entity correlations between conv3a and conv4a layers. Each violin shows the distribution of mean correlations within channels for that layer. For each channel, correlations are computed by: (1) calculating pairwise correlations between entity trajectories within a rollout, (2) averaging across all entity pairs, and (3) averaging over 10 rollouts.

4.1.1 Activation patterns over rollouts

We carried out further experiments to observe how activation patterns evolved over the course of a rollout as different entities were collected. We captured activations over the course of rollouts in both the custom maze environments and unmodified mazes, taking the mean activation in each channel at every timestep. This revealed clear patterns where activations would typically jump upwards as each key was collected, with the size of the jump decreasing after each collection. These jumps were particularly obvious in the highest-variance channels, with the effect present but not ubiquitous amongst other channels. Results of a randomly selected rollout are shown in Figure 4.3.

Figure 4.3 illustrates a recurring ‘pinching’ effect, where activations converge to similar values across entities. This is likely because the network has learned that

regardless of the entity involved, if the player is one step away from the entity then it always makes sense to go directly to that entity, as in the normal course of a rollout there is no scenario in which it can be directly beside an entity while it would be incorrect to collect that entity, except in the case of locks.

Interestingly, in the cross maze where it was possible to access all the keys from the beginning, the model would choose to target the green key first in about 12% of cases ($n=500$ rollouts). In these instances we would still see the positive jump in activations as we saw in the case of the blue key, indicating that this phenomenon is primarily tied to the number of entities collected, rather than targeting a specific entity.

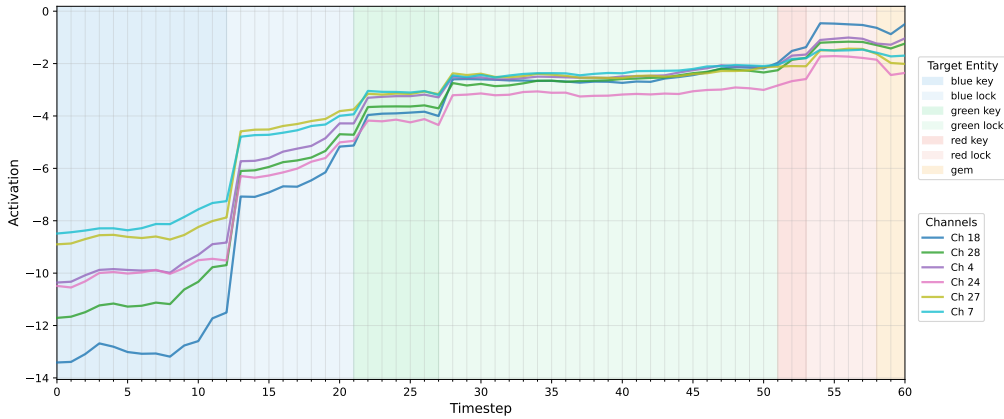


Figure 4.3: Plot showing the mean activation values captured across all timesteps in an unmodified rollout in the 6 channels that had the greatest variance over the course of the rollout. The shaded regions indicate the current ‘next target’, the entity that it would logically pursue next in solving the maze. Note the upward jumps in activations at the points between shaded regions, indicating contact with an entity.

4.2 Activation Offsets Achieve Steering

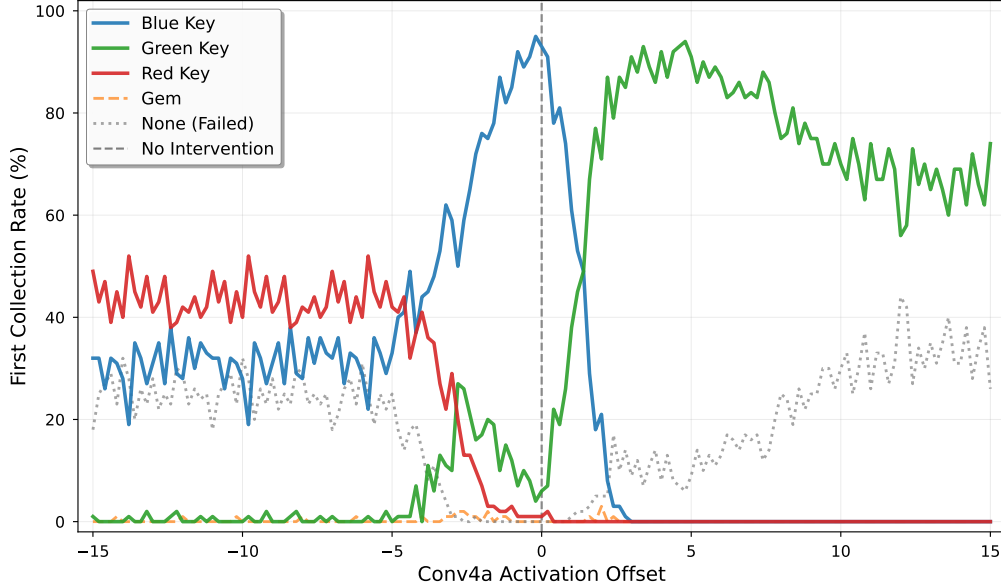


Figure 4.4: Activation offset steering results for layer conv4a at checkpoint 35k. The x-axis shows the offset value applied uniformly across all channels, while the y-axis shows the percentage of rollouts where the agent first collected each entity type. Positive offsets induce green key targeting (peaking at 94% at offset +4.8), while negative offsets shift behaviour towards red key collection (52% at offset -5.8). Blue key collection dominates near baseline (93% at offset 0.0). A run ends immediately upon contact with an entity, and is considered ‘none’ if no entity is collected within 100 timesteps.

Surprisingly, a static offset to channel activations reliably steered the agent towards different entities than what the agent would target by default. Results of rates of initial pickup over different rollouts are presented in Figure 4.4. We also observe this result on different checkpoints, with different success rates for offset values but with similar patterns. We find that steering towards different entities by adjusting mean offset works across all layers, suggesting that this is endemic to the structure of the network and not simply an isolated phenomenon.

We observe complete shifts in entity targeting. Baseline rates show $93\% \pm 5\%$ blue key targeting at offset 0.0. At positive offsets, the agent switches to targeting the green key, peaking at $94\% \pm 5\%$ at offset +4.8. At negative offsets, we observe peak red key collection of $52\% \pm 10\%$ at offset -5.8. Rates of not collecting any entity increase at the extremes, but overall collection remains fairly high, meaning important navigational functions continue to function.

Conv2a (shown in Figure 4.5) shows particularly amenable offset steering with fine grained control. Red key steering peaks at $75\% \pm 5\%$ at offset $+0.30$, illustrating clear and strong steering. There is also a surprising surge in gem collection ($12.2\% \pm 3.2\%$ at offset $+0.40$). The precision of the technique is notably high in conv2a, likely due to it being the first layer where the next target is being established in the hierarchy as illustrated by Table 4.1.

Conv1a, despite having minimal entity probe accuracy (20.0% overall), shows strong steering capabilities with a different distribution than that observed in the other layers (detailed results in Appendix B). Unlike the overlapping, nuanced patterns in later layers, conv1a exhibits discrete zones: green dominates the negative range (60-73% from -3.0 to -0.5), blue shows a sharp peak near the baseline (92% at $+0.1$), and red controls the positive range (50-75% from $+0.5$ to $+3.0$). Surprisingly we find the best gem steering in conv1a with $50\% \pm 5\%$ gem collection at $+1.9$ offset, far exceeding conv2a (12%) and conv4a (3%). This suggests conv1a interventions are able to shift the normal bias towards collecting keys first, before target integration occurs.

We also investigate whether the colours in the background have a significant impact on which entities are selected for targeting, and we find in an $n=1000$ sample that there is no statistically significant relationship between the colours of the background and the choice of entity ($p=0.21$). Figures for conv3a, conv2a, and conv1a, including more precise sweeps over smaller regions are included in Appendix B.

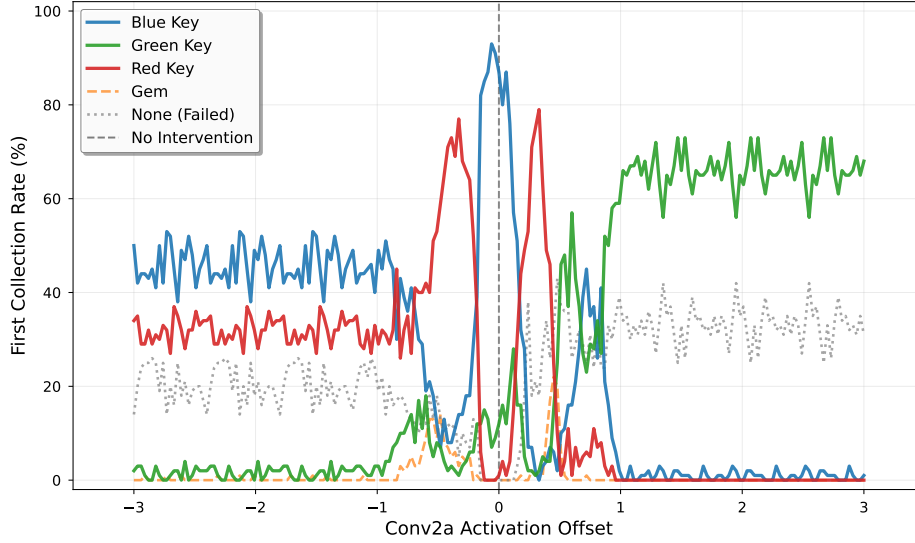


Figure 4.5: Activation offset steering results for layer conv2a at checkpoint 35k, showing fine-grained control over entity targeting. The x-axis shows offset values in the range $[-3, +3]$ with 0.03 step increments, while the y-axis shows the percentage of rollouts where the agent first collected each entity type. Conv2a exhibits particularly precise steering: red key collection peaks at $75\% \pm 5\%$ at offset $+0.30$, with a notable surge in gem collection ($12.2\% \pm 3.2\%$) at offset $+0.40$. The narrow effective ranges demonstrate that conv2a encodes entity preferences with high sensitivity to activation magnitude.

4.2.1 Quantitative Patching Results

We apply our incremental patching experiments across all channels in conv3a and conv4a and find a large number of channels which successfully redirect the agent away from a given entity. We present our results from layer conv4a in Figure 4.6. For conv3a a similar phenomenon is observed but with far more channels and many more effective patching values. Notably, different entities respond to different activation ranges within the same channels, suggesting the network uses activation magnitude to distinguish between targets.

A possible explanation for why the ranges have an upper limit beyond which they stop working is that the activation is so strong that it disrupts normal functioning of the network, meaning that the agent fails to reach the artificially activated region.

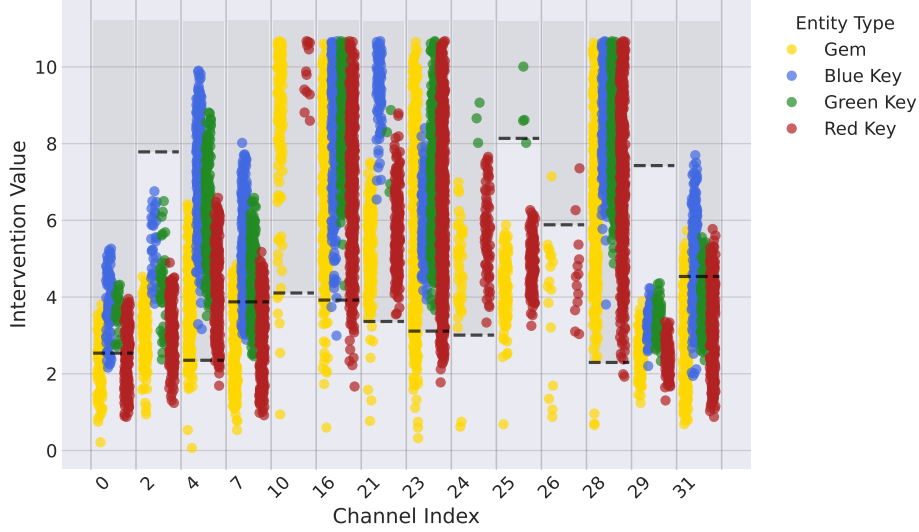


Figure 4.6: Activation patching results for layer conv4a at checkpoint 35k, showing which activation values successfully redirect the agent away from each entity type. The x-axis shows the patched activation value applied to a single spatial location in each channel, while the y-axis shows the channel index. Each coloured dot indicates a successful intervention where the agent moved towards the patched location rather than the entity. Different entities (blue key, green key, red key, gem) show distinct effective value ranges, with some channels responding to multiple entities at different activation magnitudes. Black dotted lines indicate the 99.9th percentile activation threshold for each channel, revealing that many successful interventions require super-stimulus values outside normal operating ranges.

The fact that specific value ranges seemed to correspond with specific entities reinforces the idea that the magnitude of the activation plays a real role in specifying a particular entity.

The primary value of this finding is in demonstrating that individual channels can steer the agent toward different entities at different activation strengths. We trained SAEs on these layers and found they exhibit similar patterns, suggesting this activation-range encoding is a robust strategy rather than simply the result of polysemanticity.

4.3 Using SAEs to Validate Activation Encoding

4.3.1 SAE Performance

SAEs recovered 99.6% (conv4a) and 102% (conv3a) of task performance as measured by average reward, with the slight over-recovery likely due to random variance. Recovery occurred within 40,000 training steps, with variance explained scores approaching 100% within 120,000 steps. SAEs were trained for 2 million steps to ensure convergence. This strong recovery indicates the SAEs worked as intended and could reliably validate our other results.

4.3.2 SAEs Preserve Base Model Structure

Training SAEs on conv3a and conv4a did not reveal additional interpretable structure beyond what we observed in the base model. SAE latents exhibited the same patterns of systematic activation level differences between game states, with variations in the intervention spans. This preservation across radically different representation schemes suggests the activation patterns we observe are fundamental to the operations of the network and not simply an artefact of compression.

Our primary method for using SAEs to ensure that we hadn’t simply found polysemanticity in the network was to use the same quantitative patching approach described in Section 4.2.1 applied to all of the latents in an SAE. If it were natural to encode different entities into separate channels, we would expect the SAE to learn distinct latents for each entity type. Instead, we observe broadly the same patterns: individual SAE latents successfully redirect the agent away from multiple entities, with similar activation patterns to the base model. Figures illustrating this can be observed for conv3a and conv4a SAEs in Appendix F.

Patching with Dead Latents

Many effective steering latents were ‘dead latents’ that exhibited zero activation during normal operation. We confirmed these remained inactive across 100 rollouts of typical maze-solving behaviour.

We attribute this to SAE training dynamics. These encoder-decoder pairs initially encoded useful signal during early training, but were subsequently suppressed as the L1 sparsity penalty forced lower activations. While encoder weights were driven toward near-zero values, decoder weights retained the directional structure learned when the latent was active. Artificially activating these dead latents during intervention experiments allows the preserved decoder weights to project meaningful directions into the activation space, influencing agent behaviour despite the latents never activating during normal operation.

4.4 Exploring The Role of Negative Activations

While the activation strength encoding mechanism above is clear and seems connected to increasingly negative activations over the course of an episode, it is unclear what the negative activations are doing in the network and why the activations become less negative over time.

To understand this, we analysed how negative activations changed in the convolutional layers over the course of a rollout. We observed that negative regions follow a clear pattern: regions previously explored in the maze have their activation values shift to near zero upon the collection of an entity, as shown in Figure 4.7. In this figure, channel 18 is chosen because it has the highest mean activation variance of all channels, clearly demonstrating the shifting effect.

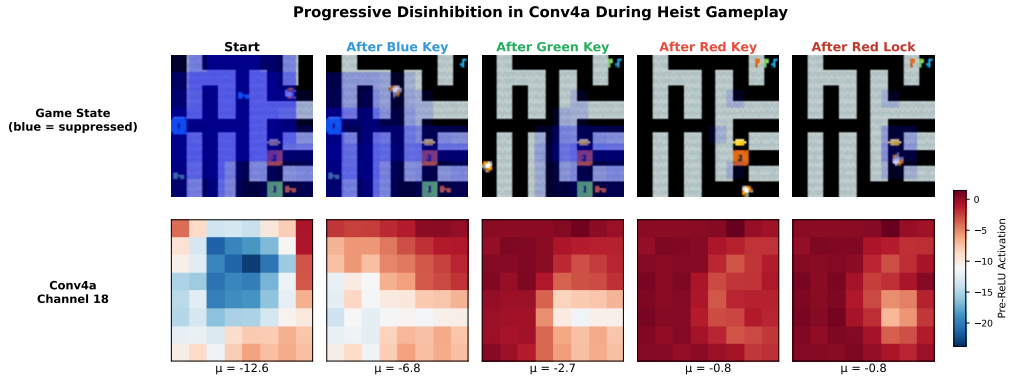


Figure 4.7: Progressive disinhibition in Conv4a during Heist gameplay. Top row: Game observations with blue overlay indicating regions of strong negative pre-ReLU activation (intensity proportional to magnitude). Bottom row: Spatial activation maps for Conv4a channel 18, with mean activation (μ) shown below each panel. At the start of an episode, activations are strongly negative across most spatial locations ($\mu = -12.6$). As the agent collects entities (keys, locks), these negative activations progressively release toward zero, with mean activation increasing to $\mu = -0.8$ by the end of the rollout. This pattern suggests the network uses negative activations to suppress representations of future objectives, with suppression lifted as each objective is completed.

Collection triggers a phase shift: the entity’s region becomes marked as empty, while regions of the maze still containing entities remaining largely negative. We observe this phenomenon across mazes and in layers conv2a, conv3a, and conv4a.

This corresponds with the jumps in mean activation observed in Figure 4.3, providing a clearer picture of what causes this phenomenon.

We test this mechanism by using patching in the convolutional layers to clamp activations in regions we do not want the agent to enter to a value of 0. This completely stops the agent from entering those regions. Tests in our cross maze environment (Figure 4.8) reveal 100% retargeting of the agent to regions that are not clamped.

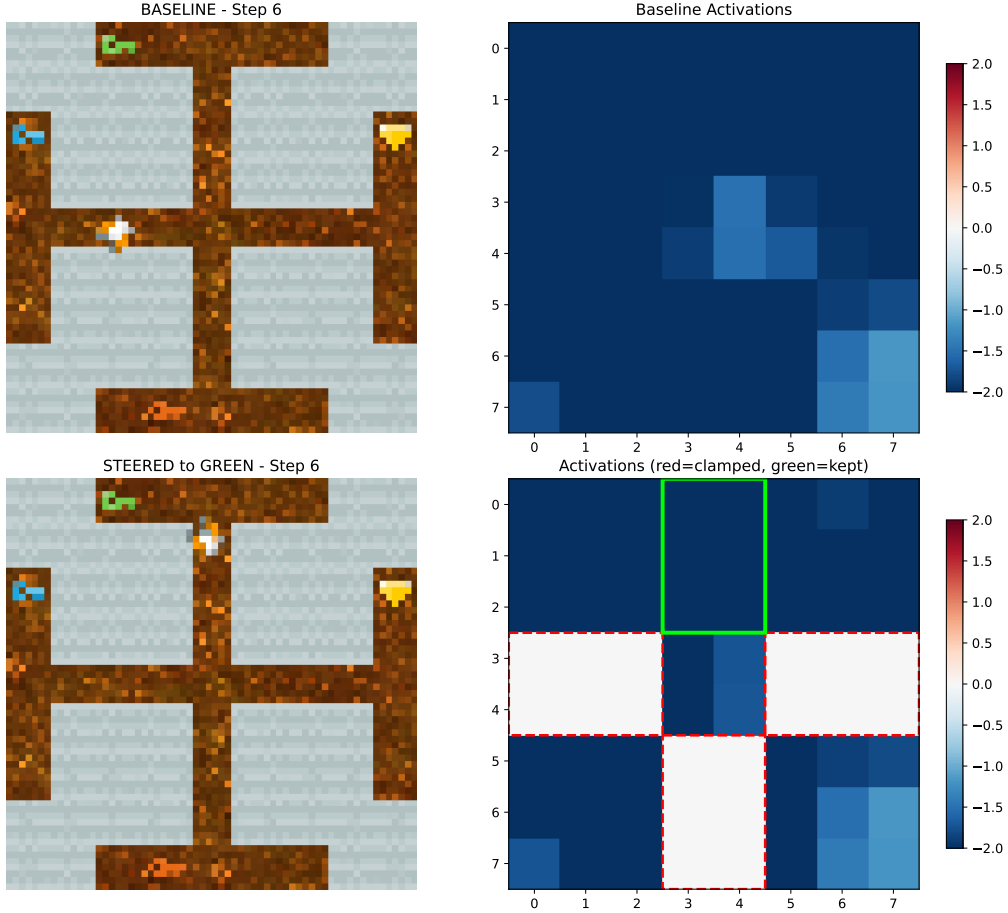


Figure 4.8: Top row: baseline behaviour where the agent moves toward the blue key (left arm). Bottom row: after clamping activations to zero in three spatial regions (red dashed boxes), the agent instead moves toward the green key in the remaining unclamped region (green box).

To verify that this mechanism operates spatially rather than on entity representations, we shuffled entity positions across maze arms and tested whether clamping still redirected the agent. With shuffled positions, clamping a region caused the

agent to avoid that direction regardless of which entity occupied it, which we refer to as repulsion. Through backward elimination, we identified a minimal set of 13 channels sufficient for position-invariant spatial repulsion: [0, 2, 3, 9, 11, 12, 14, 15, 20, 26, 29, 30, 31]. This set reliably prevents the agent from entering any clamped region regardless of which arm of the maze the entity was in. Some channels were particularly important for repelling the agent such as channel 26, which lowered the successful repulsion rate by 25%.

This causally demonstrates the spatial navigation mechanism: negative activations mark regions of interest, while near-zero activations indicate regions to avoid. This leads the model to navigate toward areas of negative activation. The entity retargeting we observe through activation offsets likely works because early stages of the game have more negative activations overall; shifting the offset mimics the activation patterns of later game stages rather than directly encoding entity identity.

4.5 Probing for Entity Representations

To determine where the network encodes information about which entity to target next, we trained linear probes on activations from each layer. We gathered 14,000 balanced samples across seven entity types (blue/green/red keys and locks, plus gem) and trained multi-class classifiers with a 90/10 train-test split. This approach reveals at which layer the network most explicitly represents the current goal.

4.5.1 Convolutional Layers Encode Entities

The probe accuracies across layers reveal that the convolutional layers dominate in encoding which entity to target, while the fully connected layers rapidly transition to translating information into actions. Table 4.1 shows whole-layer probe accuracies from the input through to the final hidden layer.

The probe accuracies illustrate two phases of information processing:

Phase 1 - Emergence (conv1a $\rightarrow$ conv3a): The whole layer probe of Conv1a performs barely above random baseline ($20.0\% \pm 2.1\%$ vs 14.3%), on everything except the green key (99.4%). However, examining individual channels (full results in Appendix D) reveals that conv1a contains highly specialised entity detectors: five channels achieve $>90\%$ accuracy on blue keys (with channel 11 reaching 100%), five channels exceed 90% on red keys (channels 9 and 14 both at 100%), and three channels on the gem (with channel 0 at 97.3%). This shows that conv1a contains specialised entity detectors, but whole-layer entity information is not yet linearly separable at this early stage, suggesting integration occurs in later layers. In contrast, conv2a achieves $96.6\% \pm 1.0\%$ accuracy across all entity types, representing a 76.6 percentage point jump. This dramatic difference shows that

Table 4.1: Whole-layer probe accuracy for predicting next target entity across network layers. Random baseline is 14.3%. Overall accuracy shown with 95% confidence intervals (n=1,400 test samples).

Layer	Overall	Blue Key	Green Key	Red Key	Gem	$\Delta \rightarrow$ Previous
conv1a	20.0% $\pm$ 2.1%	0.0%	99.4%	0.0%	0.0%	—
conv2a	96.6% $\pm$ 1.0%	97.7%	97.7%	85.6%	99.5%	+76.6%
conv3a	99.3% $\pm$ 0.4%	99.1%	99.4%	98.6%	100.0%	+2.7%
conv4a	97.6% $\pm$ 0.8%	99.1%	98.3%	97.6%	100.0%	-1.6%
fc1	87.5% $\pm$ 1.7%	95.4%	80.2%	80.3%	96.2%	-10.1%
fc2	76.8% $\pm$ 2.2%	92.6%	55.8%	75.0%	90.8%	-10.7%
fc3	20.4% $\pm$ 2.1%	23.3%	16.9%	20.2%	35.9%	-56.4%

conv2a integrates entity information into a structure from the signals in conv1a. Performance improves slightly in conv3a (99.3% $\pm$ 0.4% accuracy) before showing a slight decline at conv4a (97.6% $\pm$ 0.8%).

Phase 2 - Compression (fc1 $\rightarrow$ fc3): The clarity of entity representation decreases progressively through the fully-connected layers, dropping from 87.5% $\pm$ 1.7% (fc1) to 76.8% $\pm$ 2.2% (fc2) and finally collapsing to 20.4% $\pm$ 2.1% in fc3, barely above random. To confirm whether the information was simply being represented non-linearly, we trained MLP probes (two hidden layers with ReLU activations) on these layers. The MLP probes performed very similarly to the linear probes, suggesting the information loss is genuine rather than a limitation of linear readout.

This marks the point at which the network transforms explicit entity representations into action-oriented features. This makes sense as in final layer it has to produce only the action distribution and value estimation, which would naturally be quite distant from the entity representations themselves.

4.5.2 Directional Navigation Probes

To understand which points of the network are most involved with navigation, we trained linear probes to predict the optimal movement direction (up, right, down, left) from layer activations. We gathered 8,000 samples (2,000 per direction) using Dijkstra’s algorithm to compute optimal paths, filtering out examples where the agent was not centred in a grid cell to avoid ambiguity, and also removing examples where the agent was directly next to the target entity in order to enhance the difficulty of the problem. Results are shown in Table 4.2.

The directional probes reveal a significantly different structure than the next tar-

Table 4.2: Whole-layer probe accuracy for predicting optimal movement direction across network layers. Random baseline is 25%. Overall accuracy shown with 95% confidence intervals (n=1,600 test samples).

Layer	Overall	Up	Right	Down	Left	$\Delta \rightarrow$ Previous
conv1a	25.1% $\pm$ 2.1%	100.0%	0.3%	0.0%	0.0%	—
conv2a	42.7% $\pm$ 2.4%	44.5%	61.5%	5.5%	59.2%	+17.6%
conv3a	63.6% $\pm$ 2.4%	61.0%	66.8%	69.0%	57.8%	+20.9%
conv4a	51.7% $\pm$ 2.5%	51.2%	48.8%	51.0%	55.8%	-11.9%
fc1	62.9% $\pm$ 2.4%	60.5%	63.5%	61.5%	66.2%	+11.2%
fc2	67.3% $\pm$ 2.3%	64.5%	66.2%	69.2%	69.2%	+4.4%
fc3	21.1% $\pm$ 2.0%	37.0%	16.8%	21.0%	9.8%	-46.2%

get entity probes. Conv2a achieves only 42.7% $\pm$ 2.4% accuracy (barely above the 25% baseline), with highly unbalanced performance, achieving near-zero accuracy on downward movement (5.5%) despite reasonable performance on horizontal directions. The best convolutional layer performance is observed at conv3a (63.6% $\pm$ 2.4%), before showing a somewhat surprising drop at conv4a to 51.7% $\pm$ 2.5%.

Crucially, unlike entity information which degrades in fully-connected layers, directional information *recovers and improves* through the FC layers. FC1 recovers to 62.9% $\pm$ 2.4%, and fc2 achieves the highest accuracy overall at 67.3% $\pm$ 2.3%, with balanced performance across all four directions. This suggests that spatial navigation information is actively refined and strengthened as the network prepares for action selection, rather than being compressed away.

The surprising conv4a performance drop (63.6% $\rightarrow$ 51.7%) diverges from the pattern observed in the next target entity probes, which retain most of their accuracy at this layer (99.3% $\rightarrow$ 97.6%). This suggests conv4a functions as a bridge in converting the high level spatial information into a structure coherent to the fully connected layers that temporarily disrupts the linear separability of directional information.

4.5.3 Comparing Entity and Directional Information Flow

The contrasting progressions reveal a two-stage processing architecture. Entity information contains specialist detectors even at conv1a and peaks at conv3a (99.3% $\pm$ 0.4%), while directional information is entirely absent at conv1a (25.1% $\pm$ 2.1%, at random baseline) and peaks much later at fc2 (67.3% $\pm$ 2.3%). This indicates that the network implements a hierarchical processing strategy:

Stage 1 - Goal Identification (conv2a–conv4a): Entity information emerges sharply from conv1a (which contains specialist entity detectors despite poor whole-layer performance) through conv2a ($96.6\% \pm 1.0\%$) and peaks at conv3a ($99.3\% \pm 0.4\%$), remaining extremely high at conv4a at $97.6\% \pm 0.8\%$. At this stage, directional information remains relatively weak ($63.6\% \pm 2.4\%$ at conv3a), suggesting the network prioritises identifying the correct goal before planning how to reach it. Notably, conv1a contains no directional information whatsoever ($25.1\% \pm 2.1\%$, at random baseline), contrasting with its entity specialist channels, indicating that directional processing emerges later in the network.

Stage 2 - Navigation Planning (fc1–fc2): As entity information compresses ($99.3\% \rightarrow 20.4\%$), directional information strengthens ($63.6\% \rightarrow 67.3\%$). The network transforms explicit entity representations into spatial navigation commands, with peak directional accuracy at fc2 ($67.3\% \pm 2.3\%$) suggesting motion-oriented features directly relevant to action selection. Both types of information then collapse at fc3 (entity: $20.4\% \pm 2.1\%$, direction: $21.1\% \pm 2.0\%$), as they are compressed into the final action distribution and value estimates.

This functional separation explains key findings from our intervention experiments. The near-perfect entity encoding at conv3a explains why patching and ablation interventions were most effective at this depth: we modified explicit goal representations before their transformation into motor commands. Conversely, the distributed nature of directional information across multiple layers explains why navigation remained largely functional even under heavy ablation, making it more robust to localised disruption than the concentrated entity representations.

4.6 Ablation Studies

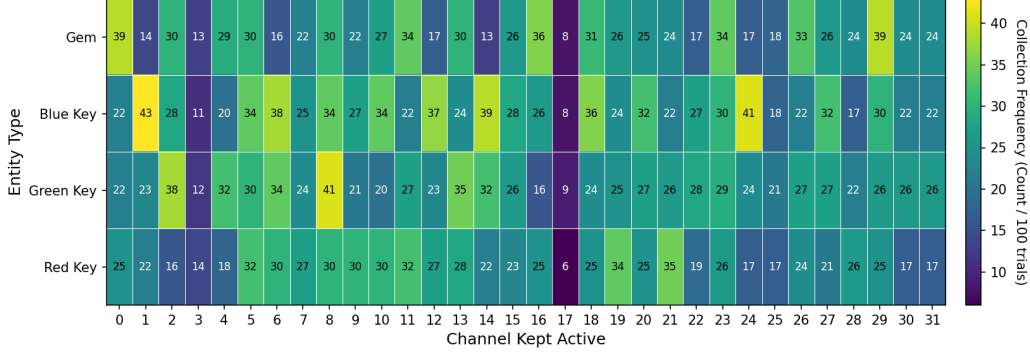


Figure 4.9: Heatmap showing entity collection counts when only a single channel is active (all others ablated). The vertical axis shows different entity types (gem, blue key, green key, red key), while the horizontal axis shows channel indices. Brighter colours (yellow) indicate higher collection success rates.

Our ablation analysis reveals that the ability to navigate is surprisingly redundantly encoded throughout the network. The heatmap in Figure 4.9 shows the results in full. We observe that all channels have some success reaching entities, with the best performing channels achieving success rates greater than 40% for certain entities, while others show lower than 10% success rates. Some channels show similar performance across entities, while others are significantly better at particular entities.

We performed extensive targeted ablation experiments to better understand the functional roles of different channel groups, including ablating intervention spans (the activation ranges where patching successfully redirected the agent), navigation channels (those with highest independent collection rates), and infrastructure channels (those whose removal most impacted collection). However, these experiments yielded complex and often counterintuitive results that did not clearly illuminate the network’s core mechanisms. Full results are provided in Appendix G.

The most interpretable finding was that intervention spans alone are not sufficient to drive entity collection. When we preserved only the intervention spans for each entity while ablating everything else, performance fell below random-control baselines: 31.8% for gem versus 44.1% random, 61.8% for blue key versus 74.9% random, and similar patterns for other entities. This confirms that the intervention spans encode entity-relevant information but depend on other navigation circuitry to be effective; they represent one component of a distributed system rather than a self-contained targeting mechanism.

4.7 Summary of Key Findings

Our investigation reveals four key insights into how this network implements goal-directed behaviour:

1. Activation Patterns Track Game State. Parallel rollouts show consistent activation offsets between runs targeting different entities (correlation 0.931 in conv4a), and complete behavioural redirection is achievable through uniform offset adjustments (93% blue key at baseline to 94% green key at +4.8 offset). The network’s strong baseline preference for the blue key supports our hypothesis regarding a learned hierarchical structure. This pattern persists across layers and through SAE decomposition, suggesting it represents a fundamental computational strategy rather than an architectural artefact.

2. Negative Activations Provide Spatial Gating. As entities are collected, their associated regions shift from negative to near-zero, progressively disinhibiting the spatial representation. Clamping activations to zero in any region completely prevents the agent from entering it, confirming the causal role of this gating mechanism. This spatial gating provides the foundation for the offset steering effect: early game stages have more negative activations overall.

3. Two-Phase Processing Architecture Separates Goal Selection from Navigation. Entity determination occurs in convolutional layers (peaking at 99.3% probe accuracy in conv3a) while navigation planning happens in fully-connected layers (peaking at 67.3% directional accuracy in fc2). Conv4a serves as a critical bridge, maintaining entity information while reformatting spatial representations. This separation explains why patching interventions succeed at conv3a/4a: we intervene precisely where goals crystallise, before transformation into distributed action plans.

4. Layer Depth Determines Intervention Character. Despite minimal probe accuracy (20.0%), conv1a shows strong but chaotic patching effects with discrete entity zones and uniquely achieves 50% gem steering, far exceeding later layers. Conv2a offers the finest control with ± 0.5 offset precision, representing the optimal intervention point where representations first integrate. Conv3a/4a provide stable but coarser manipulation of crystallised representations. This progression from raw detection through integration to crystallisation defines both the network’s computational strategy and our ability to control it.

Chapter 5

Discussion

5.1 Interpretation of Results

The literature review identified three gaps in RL interpretability research (Section 2.6): how agents represent and switch between multiple sequential objectives, whether SAEs disentangle representations in RL CNNs as effectively as in supervised vision models, and whether activation patterns beyond simple feature presence carry goal-relevant information. Our findings address these gaps through the discovery of a spatial gating mechanism where negative activations mark regions of current interest, with systematic shifts tracking task progress.

5.1.1 Activation Patterns and Spatial Gating

Our findings in Section 4.1, 4.2, and 4.4 reveal that the network tracks game state through systematic changes in activation patterns. Rather than requiring separate channels for each entity type, the same channels are active throughout, but at different activation levels depending on game progress.

The ability to completely redirect agent behaviour through uniform activation offsets (93% blue key collection shifting to 94% green key collection in conv4a) suggests this is central to the network’s learned algorithm. This suggests that interventions on neural networks need to consider not just which channels to modify, but how activation levels interact with correlations in state the model has learned. The effectiveness in taking advantage of this correlation in directly affecting behaviour is one of the most surprising results in our research, and presents some concerns on the effects of confounders in causal methods in ML.

This contrasts with (Mini et al., 2023) (discussed in Section 2.5), who found dedicated ‘cheese channels’ that could be individually ablated to retarget the network. In the multi-objective setting, the network reuses the same channels across entities

rather than dedicating separate channels to each, with activation levels encoding which entity is currently relevant. This suggests that representational strategies may differ substantially between single-goal and multi-goal environments.

One challenge in making this discovery was that many of the tools typically used in mechanistic interpretability were not designed to expose temporal patterns in activation levels. SAEs and probes successfully identified which channels responded to which entities, but did not reveal how activation levels shifted systematically over the course of a rollout. This differs from the superposition described by (Elhage, Hume, et al., 2022) (Section 2.2.2), where features are compressed into overlapping directions because they rarely co-occur in training data. In our case, all entities are simultaneously visible in the maze; the network is not compressing features that don't co-occur, but tracking which of several co-occurring features is currently behaviourally relevant through variation in activation strength.

5.1.2 Two-Phase Information Processing Architecture

The probe results reveal why CNN-FC hybrid architectures are well suited for goal-directed navigation. CNNs capture and classify spatial relationships necessary for identifying 'what is where', while fully-connected layers performed the task of integrating this information into coherent actions. Following (Alain and Bengio, 2018), we use linear probes to assess where information crystallises in the network. Our finding that entity information peaks at conv3a (99.3%) before collapsing in FC layers aligns with their observation that intermediate layers often contain the most linearly separable representations.

The inverse relationship between entity and directional information across layers suggests the network may perform active transformation rather than passive feature extraction. Conv4a's role as a bridge between the spatial and fully-connected layers is illustrated by the non-monotonic increase in the linear separability of the direction in which to go from conv3a to fc2. This implies that the linear separability drops due to an active transformation taking place in the structure of the information in conv4a. Future work in understanding this transformation would be valuable in establishing how RL agents transform visual information into motor output.

This crystallisation of goal commitment occurs at conv3a/conv4a. Our patching experiments target the precise point where the network commits to its goal, before that commitment disperses into the distributed representations necessary for action selection. Interventions earlier would affect raw perception; interventions later would fight against already-distributed action plans.

5.1.3 Solution Multiplicity and Representational Drift

Intervention positions changed completely across checkpoints even after convergence. Despite stable performance, the network continuously explored different encoding schemes, likely driven by training on infinite maze variations. The spatial gating pattern appears consistently, but which specific channels implement it drifts across checkpoints.

This suggests the network exists on a ‘valley floor’ of equally viable solutions, constantly reorganizing internal representations while preserving behavioural performance. One checkpoint (step 30001) showed no steerable channels for any entity, the only such case in 60 checkpoints tested, suggesting this configuration may be less optimal than those with stronger intervention structure.

Training required unlimited procedural generation; attempts with 200 or 500 fixed mazes failed to train successfully, indicating the diversity was necessary for this architecture.

5.1.4 Environmental Conditions That May Encourage This Strategy

Two features of Heist may have encouraged the encoding strategies we observe. First, the same navigation mechanism is used repeatedly across stages, making resource sharing optimal. Second, there is strong pressure to correctly distinguish between entities: targeting the wrong entity almost certainly causes failure. These pressures may explain why the network developed systematic activation-level shifts to delineate targets, though we cannot confirm this was the only viable solution.

5.1.5 How The Network Selects Targets

Modifying a single channel in a single layer is sufficient to alter behaviour, revealing redundancy in the steering mechanism. Mean activations remain well below zero for most channels during standard rollouts (Figure C.1).

Clamping experiments confirm this mechanism: setting activations to zero in specific regions completely prevents the agent from entering them, regardless of which entity occupies that region. While positive activation patching can also redirect the agent, the mechanism by which this works is less clear, since positive activations should indicate regions to avoid under the spatial gating hypothesis. As entities are collected, fewer regions require suppression, explaining the systematic increase in mean activation values over the course of a rollout.

The activation-magnitude patterns we observe in steering experiments appear to be a downstream correlation rather than the causal selection mechanism itself. Because spatial gating produces predictable magnitude shifts as each stage clears

(fewer negative regions = higher mean activation), the network has learned deep statistical correlations between overall activation magnitude and game progress. Offset steering exploits these correlations rather than directly manipulating the selection mechanism. This distinction is important: successful steering does not imply we have identified the causal mechanism, only that we have found an exploitable correlation.

Past conv1a, the network does not rely on specialised channels for target selection. Channels in conv2a, conv3a, and conv4a play critical roles for multiple entities, with target selection determined by overall activation range rather than channel identity. Some specialisation exists in conv4a (channels 24, 25, 26 focus on red key and gem in Figure 4.6), but this is the exception rather than the rule.

5.1.6 Ablation Studies

The ablation results (Section 4.6) show navigation is redundantly encoded: many channels retain some ability to reach entities when all others are masked (Figure 4.9). Results were often counterintuitive, with targeted ablations sometimes improving performance on non-targeted entities. This complexity likely reflects the shared circuitry serving multiple goals in this environment.

5.2 Theoretical Implications

The spatial gating mechanism we observe suggests this network employs more nuanced encoding strategies than simple feature presence/absence. Rather than dedicating separate channels to each goal, it uses negative activations to mark spatial regions of interest, with the same channels active across different game states. Whether similar strategies emerge in other architectures remains an open question.

As noted above, this mechanism was discovered through mean activation analysis rather than sophisticated interpretability tools, suggesting that MI approaches focused on feature identification may miss organisational principles encoded in activation dynamics.

For RL interpretability specifically, techniques successful in LLMs or simpler environments may need adaptation in complex, interactive settings. The interplay between learned environmental structure (sequential objectives) and emergent network representations is a key area for future theoretical development.

5.3 Practical Applications

The precision of offset steering (± 0.5 at conv2a is sufficient) suggests a lightweight control mechanism requiring no architectural changes or retraining. However,

this technique exploits correlations between activation levels and game state that emerged from Heist’s particular sequential structure. Other models would need to develop analogous correlations for similar interventions to work, and there is no guarantee they would. Detecting such patterns is straightforward (tracking mean activations over rollouts), but finding exploitable correlations in other settings should not be assumed.

The two-phase processing architecture provides a practical debugging framework: target convolutional layers for goal selection issues, FC layers for navigation and planning problems. Representations are most malleable early in the network where they first integrate low-level information, providing an optimal control point before entity selection disperses into distributed action plans.

5.4 Methodological Insights

While significant effort was invested in training and analysing SAEs, the primary insights stemmed from tracking how activation levels changed over the course of rollouts. Initially, apparent polysemanticity in the channels motivated the use of SAEs to disentangle features. However, the network was not truly polysemantic in the conventional sense; rather, it recycled circuits by creating unique conditions through systematic shifts in activation ranges. What appeared to be polysemanticity was instead a structured encoding strategy.

For future interpretability research, the T-maze experimental design (Figure 3.2) proved valuable by allowing isolation of key variables, and analysing activations over complete rollouts provided far more insight than examining individual observations in isolation.

Chapter 6

Conclusion

6.1 Summary of the Research

This work investigated how deep reinforcement learning agents internally represent and pursue sequential goals in the procedurally generated Procgen Heist environment. Using mechanistic interpretability techniques such as linear probes, activation interventions, parallel rollout analysis, and SAEs, we investigated the core mechanism behind how our network learned to encode goal structures in this multi-objective task and whether interpretability tools developed for language models translate to RL agents.

6.2 Addressing Research Questions and Hypotheses

We now revisit the original research questions and hypotheses from Chapter 1 and summarise the extent to which they have been addressed.

6.2.1 Hypothesis Testing

While some hypotheses were confirmed, the discovery of spatial gating was unexpected. SAEs did not disentangle features as hypothesised, but instead preserved the same activation patterns, validating that these are fundamental to the network’s operation rather than artefacts of polysemanticity.

Table 6.1: Hypothesis outcomes with key supporting evidence.

H#	Hypothesis	Outcome	Key Evidence
H1	Dedicated entity channels	Partial	Conv1a specialised, conv2-4a shared across entities
H2	SAEs disentangle features	Rejected	SAEs preserve activation-range encoding
H3	Hierarchical preferences	Confirmed	93% blue key collection at baseline (cross maze)
H4	SAEs successfully recover performance	Confirmed	99.6-102% recovery
H5	Layer specialisation	Confirmed	Entity $\rightarrow$ direction transition proven

6.2.2 Research Questions

6.3 Key Findings and Contributions

The network uses spatial gating to track game state: negative activations mark regions of interest, shifting systematically as objectives are completed. Clamping experiments confirm this causally, with 100% retargeting when regions are clamped to zero. Parallel rollout experiments show high correlation ($r=0.931$ in conv4a) between activation levels and game stage, reflecting learned statistical correlations downstream of spatial gating. Offset steering exploits these correlations to redirect behaviour (93% blue key $\rightarrow$ 94% green key), though this manipulation targets the correlation rather than the causal selection mechanism itself.

We identify a two-phase processing architecture. Entity information crystallises to 99.3% probe accuracy in conv3a before collapsing to 20.4% in fc3, while directional information strengthens to 67.3% in fc2. Convolutional layers determine what to target; fully-connected layers determine how to reach it. This explains why interventions in convolutional layers succeed: they modify goal selection before it disperses into navigational planning.

SAEs preserved rather than decomposed the activation-range encoding patterns, validating that this mechanism is fundamental to the network rather than an artefact of polysematicity. However, SAEs provided limited additional insight beyond what simpler methods revealed. This mechanism was discovered through mean activation analysis, not sophisticated interpretability tools, highlighting a gap in current MI approaches that focus on feature identification rather than temporal patterns in activation levels.

Table 6.2: Research questions and their answers based on experimental evidence.

RQ	Question	Answer
RQ1	How are goals encoded?	Via spatial gating where negative activations mark regions of interest (Section 4.1, 4.4)
RQ2	How is the current target determined?	Two-phase: detection at conv1a $\rightarrow$ integration at conv2a-conv4a (Table 4.1)
RQ3	Can SAEs clarify feature roles?	Yes, but only providing validation (Section 4.3.2)
RQ4	Dedicated vs general channels?	Conv1a specialised conv2a-conv4a generalised (Appendix D)
RQ5	Why different steering success rates across entities?	Gem spans in 31/32 channels vs 14 for keys may explain differential steering (Section 4.2.1)
RQ6	Do layers have specialised roles?	Yes: entity detection in conv layer, navigation in fc2 (Tables 4.1, 4.2)

6.4 Limitations of the Study

This research, while providing several insights, is subject to certain limitations:

Our work is currently limited by the fact that we test only on a single environment, as there are not many environments that have similar qualities such as the extreme generalisation encouraged by Procgen, and the staged multi-goal configuration in Heist. To address this, future work could attempt this with MiniGrid environments, though a custom configuration that can establish similar diversity may be needed.

It is also unclear to what extent the spatial gating phenomenon generalises to other architectures, and we plan to test this in future work.

Our training methodology differs from standard practice in using unlimited procedurally generated environments rather than the typical 200-500 set of environments one would set. While this was necessary for successful training with our compressed architecture, it may affect the generalisation of our findings to models trained under standard Procgen protocols. The unlimited environment diversity likely also contributes to the representational drift we observe, as the network

continually trains on new environmental distributions.

We also do not yet fully characterise the function of the intervention spans, or if they translate to an actual unique role within the network’s natural functioning. We plan to address this directly through a thorough application of probes that are trained to detect which entity is currently the next entity in order of pursuit by the network. Early work shows that the later convolutional layers do this very successfully, as well as the first fully connected layer. However, to truly understand the role of the intervention spans and navigation channels, it may be necessary to train probes on every channel and every row in the fully connected layer to determine their function.

Intervention experiments, by their nature, can push the network into off-distribution activation states. While this is useful for causal testing, the agent’s behaviour under such strong artificial interventions do not represent its typical behaviour. We do not yet have a clear picture of precise circuitry under normal operating conditions. Additionally, the patching interventions were typically only applied with positive activations, but the mean channel activation analysis shown in Figure 4.3 reveals that for many channels the activations of the network are mostly negative. This means that a large portion of their activity is not being captured accurately in the interventions. We believe future work in understanding the role of the negative activations in the convolutional channels would lead to fruitful insights.

Our approach of simply intervening in every channel with 400 different activation strengths is also an approach that would need to be approached more carefully, due to cost constraints, in larger models.

6.5 Future Work

The discovery of spatial gating suggests examining whether similar mechanisms exist in other architectures and domains, particularly in transformer-based models where activation patterns might track different modes of operation.

Testing this mechanism across different RL environments, especially those with hierarchical goal structures or mode-switching requirements, could reveal whether spatial gating is a general principle of efficient neural computation under resource constraints.

Revisiting other ‘well-understood’ models with basic statistical approaches could uncover overlooked organisational principles, particularly in systems previously deemed too polysemantic for interpretation.

The precise control achievable through small activation offsets (± 0.5 at conv2a) warrants investigation into whether training procedures could be designed to encourage such structures, potentially yielding models that are interpretable and

controllable by design rather than post-hoc analysis.

Finally, the two-phase architecture’s clear separation between ‘what to target’ and ‘how to navigate’ suggests applications in hybrid systems where interpretable goal selection could be combined with powerful but opaque execution modules, potentially offering a path toward more trustworthy autonomous systems.

6.6 Final Remarks

This thesis provides a concrete mechanistic account of goal representation in an RL agent through spatial gating, where negative activations mark regions of current interest. The discovery that this mechanism emerged from simple analysis rather than sophisticated tools suggests that current MI approaches may overlook organisational principles encoded in activation dynamics.

Whether spatial gating generalises beyond this specific setting remains an open question, but the representational challenge it addresses, reusing circuits across sequential objectives, is common to many goal-directed systems.

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Appendix A

Model Architecture Details

ImpalaCNN architecture:

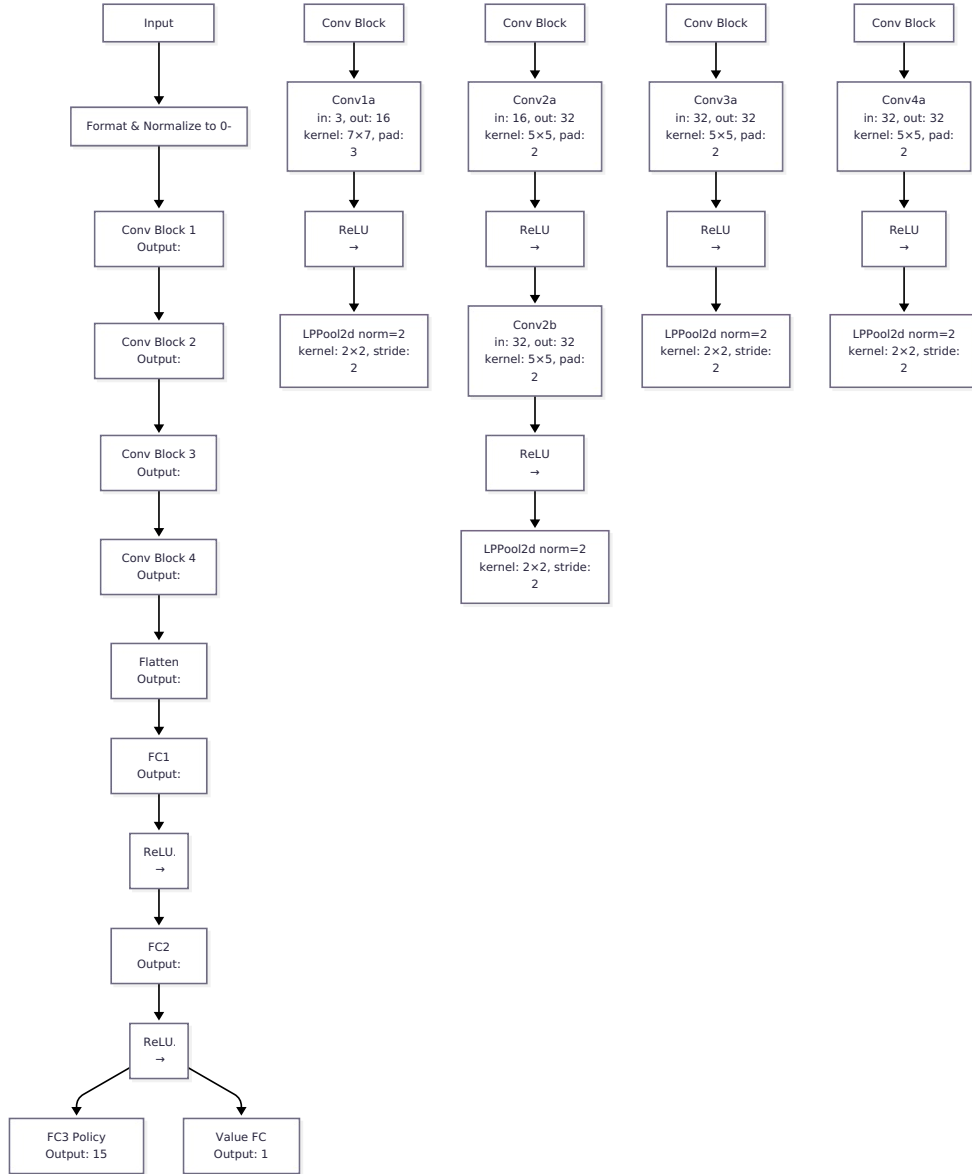


Figure A.1: The convolutional neural network architecture used in our experiments. We use a modified version of the Impala CNN with 5 convolutional layers rather than the original 15 layers, which provides better interpretability without compromising performance.

Appendix B

Activation Offset Results

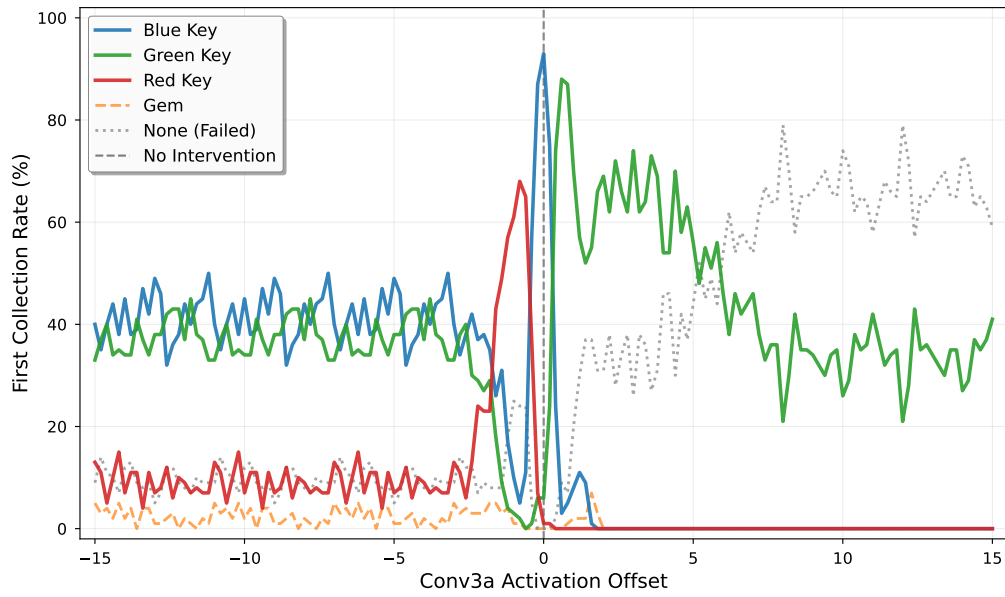


Figure B.1: Full range activation offset sweep for conv3a layer at checkpoint 35001. Each subplot shows entity collection success rates across different activation offset values for a specific channel. This wide-range sweep reveals the broad activation ranges over which different channels can successfully steer the agent toward specific entities.

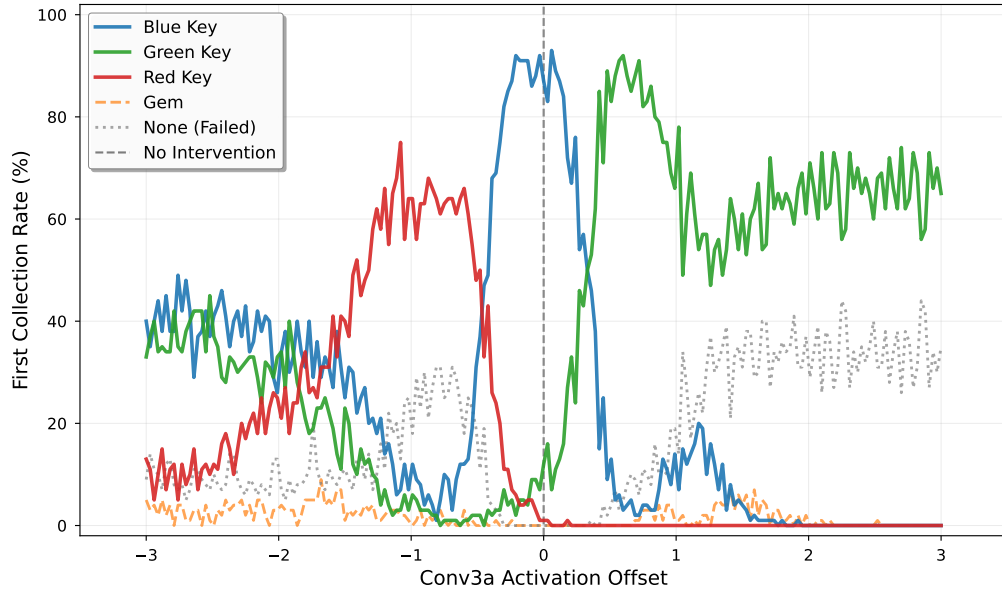


Figure B.2: Narrow range activation offset sweep for conv3a layer at checkpoint 35001, focusing on offset values from -3 to +3 with step size 0.03. This fine-grained sweep provides higher resolution detail of entity collection rates in the critical activation range where steering behaviour transitions occur.

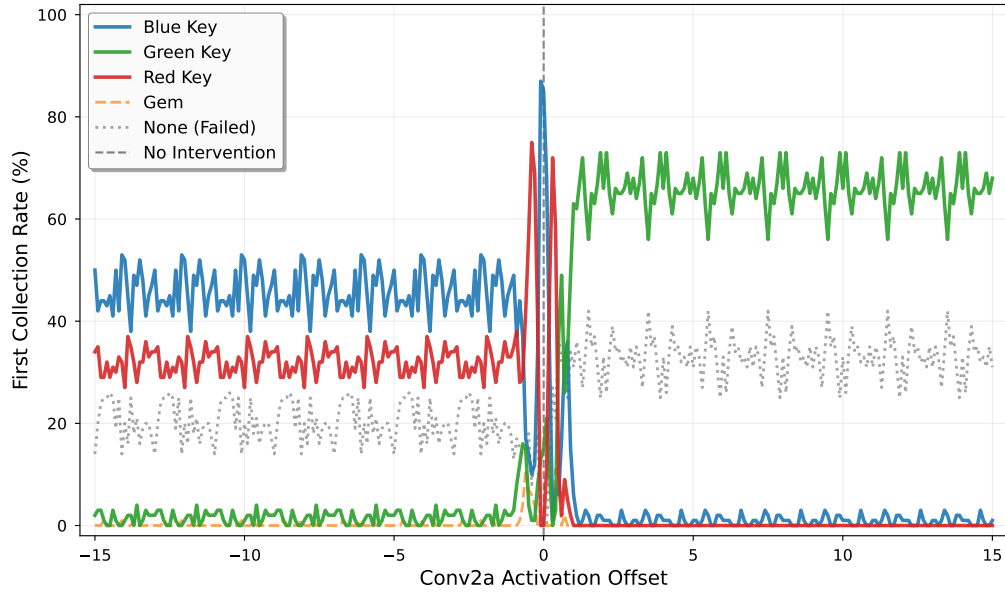


Figure B.3: Wide range activation offset sweep for conv2a layer at checkpoint 35001, testing offset values from -15 to +15 with step size 0.1. This broader sweep examines the extent to which earlier convolutional layers can be steered across a wide range of activation offsets.

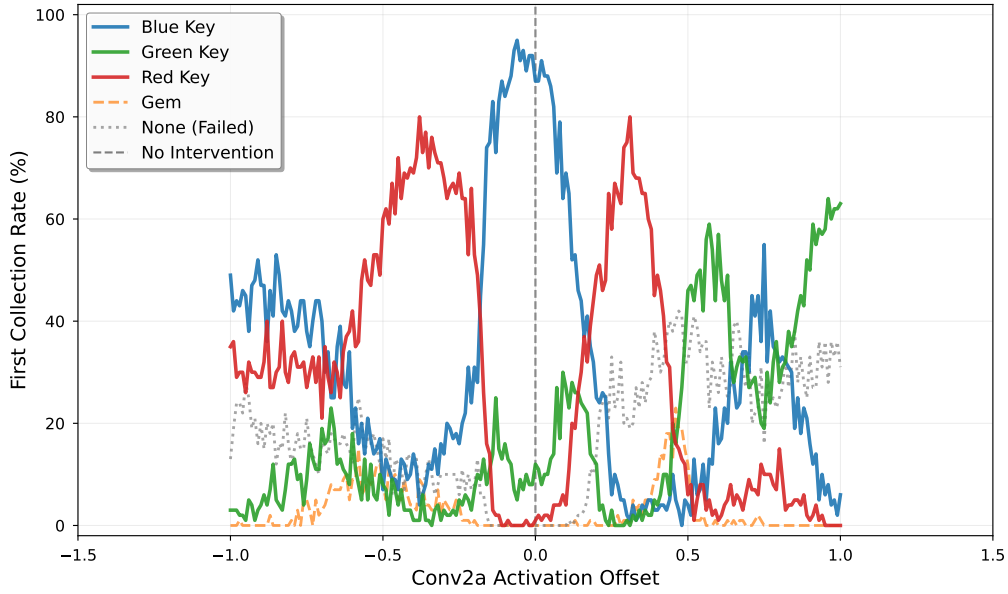


Figure B.4: Narrow range activation offset sweep for conv2a layer at checkpoint 35001, focusing on offset values from -1 to +1 with step size 0.01. This high-resolution sweep examines fine-grained steering behaviour in the conv2a layer’s most sensitive activation range.

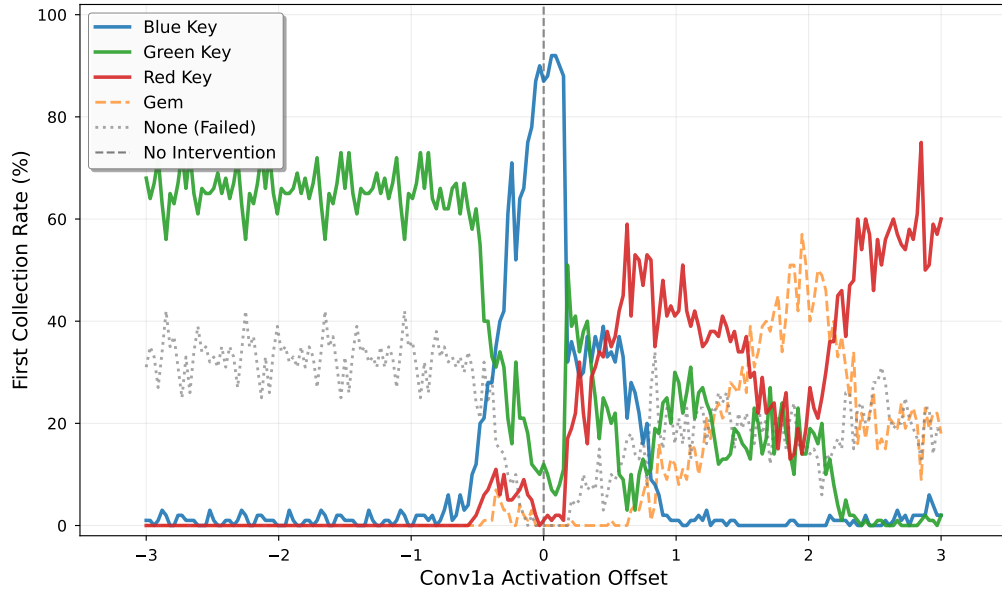


Figure B.5: Activation offset sweep for conv1a layer at checkpoint 35001, testing offset values from -3 to +3 with step size 0.03. This sweep examines steering capabilities in the earliest convolutional layer, which operates closest to the raw input observations.

Appendix C

Mean Activation Per Channel During Rollout

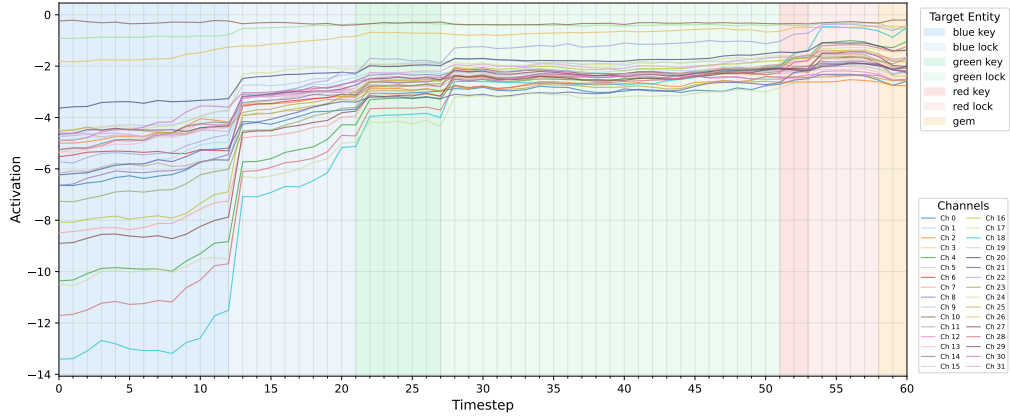


Figure C.1: Figure showing the mean activation per channel across all channels in the same rollout as presented in Figure 4.3. We see that the pattern of increasing activation strength over the course of an episode with sharp jumps at the collection of different entities is shared across many but not all channels, with a few showing very consistent activation strengths from beginning to end. In general mean activations remain below 0 for all channels, indicating that a great deal of the neurons across channels are negative a majority of the time.

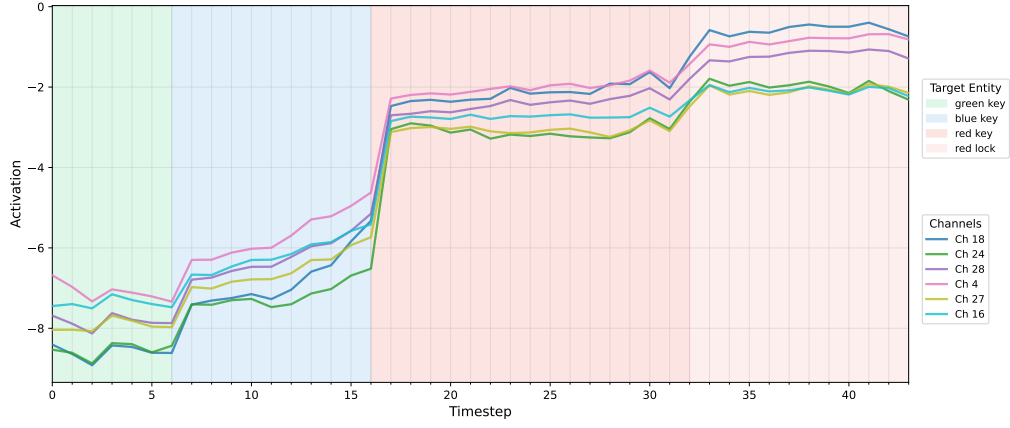


Figure C.2: Plot showing the mean activation values captured across all timesteps in an unmodified rollout in the 6 channels that had the greatest variance over the course of the rollout. The shaded regions indicate the current ‘next target’, the entity that it would logically pursue next in solving the maze. In this rollout in the cross maze environment, the agent targeted the green key first. A similar uptick in positive values can be observed at the moment of green key collection, though the larger uptick in mean activations typically associated with the blue key can be observed at the moment of blue key collection as well.

Appendix D

Channel-Level Probe Results

This section presents detailed per-channel probe accuracies for multi-class entity classification across convolutional layers. All probes were trained using logistic regression on individual channel activations to predict the next entity the agent will navigate toward (7-way classification: blue key, green key, red key, blue lock, green lock, red lock, gem). Lock entities are omitted from these tables for brevity, though they were included in the overall accuracy calculations. Random baseline performance is 14.3%.

D.0.1 Conv1a Layer - All Channels

Table D.1 shows probe results for all 16 channels in the Conv1a layer. This layer exhibits the highest variance in channel performance, with the best channel (Ch0) achieving 97.1% overall accuracy while the worst (Ch5) achieves only 14.7%. Three channels (Ch0, Ch11, Ch12) demonstrate strong generalist performance across all entity types, suggesting they encode rich multi-entity representations.

Table D.1: Conv1a layer per-channel probe accuracies. All 16 channels shown. Bold indicates highest accuracy in each column.

Channel	Overall	Blue Key	Green Key	Red Key	Gem
0	97.1%	97.7%	94.8%	98.1%	97.3%
1	17.7%	0.0%	0.0%	79.3%	45.1%
2	65.1%	90.7%	14.5%	94.7%	90.2%
3	20.3%	0.0%	3.5%	0.0%	50.5%
4	17.2%	98.1%	0.0%	0.0%	4.4%
5	14.7%	0.0%	0.0%	0.0%	0.0%
6	21.9%	28.4%	0.0%	54.8%	0.0%
7	16.1%	0.0%	0.0%	94.7%	0.0%
8	21.1%	0.0%	61.6%	0.0%	53.8%
9	14.9%	0.0%	0.0%	100.0%	0.0%
10	17.1%	0.0%	43.6%	76.0%	0.0%
11	94.7%	100.0%	98.3%	90.4%	88.6%
12	91.6%	99.5%	94.8%	88.5%	89.7%
13	20.3%	0.0%	0.0%	54.3%	0.0%
14	15.2%	2.3%	0.0%	100.0%	0.0%
15	17.7%	0.0%	0.0%	0.0%	48.4%

D.0.2 Conv2a Layer - Top 10 Channels

Table D.2 presents the top 10 performing channels in Conv2a (out of 32 total). This layer shows dramatically improved performance compared to Conv1a, with the best channel (Ch4) achieving near-perfect 99.4% accuracy. The top channel achieves perfect accuracy on three entity types (blue lock, green lock, red lock, gem), suggesting highly disentangled representations.

Table D.2: Conv2a layer top 10 channels by overall accuracy (out of 32 total channels). Bold indicates highest accuracy in each column.

Channel	Overall	Blue Key	Green Key	Red Key	Gem
4	99.4%	98.1%	98.8%	99.0%	100.0%
27	98.9%	98.6%	99.4%	96.6%	100.0%
1	95.2%	98.6%	96.5%	94.2%	98.9%
25	93.9%	97.2%	94.8%	88.0%	96.2%
14	90.4%	88.8%	94.2%	83.2%	100.0%
31	90.2%	92.6%	88.4%	86.1%	95.1%
5	88.4%	92.1%	86.1%	75.0%	96.7%
29	87.6%	95.4%	89.5%	77.9%	96.2%
18	86.4%	88.4%	84.3%	77.4%	100.0%
16	85.5%	95.8%	77.9%	70.7%	91.9%

D.0.3 Conv3a Layer - Top 10 Channels

Table D.3 shows the top 10 channels in Conv3a (out of 32 total). Interestingly, this layer shows lower per-channel performance than Conv2a (best: 82.9% vs 99.4%), despite whole-layer probes achieving 99.3% accuracy. This suggests Conv3a relies more heavily on distributed representations across multiple channels rather than individual specialist channels.

Table D.3: Conv3a layer top 10 channels by overall accuracy (out of 32 total channels). Bold indicates highest accuracy in each column.

Channel	Overall	Blue Key	Green Key	Red Key	Gem
19	82.9%	89.8%	79.1%	80.8%	90.2%
20	82.7%	77.2%	79.1%	80.3%	88.6%
9	78.6%	77.7%	79.1%	76.4%	90.2%
4	78.4%	83.7%	75.0%	72.1%	83.7%
6	76.4%	84.7%	72.1%	73.1%	85.3%
25	76.2%	79.1%	72.1%	62.5%	97.8%
23	75.6%	87.9%	66.9%	71.2%	85.3%
2	74.4%	77.7%	66.3%	74.0%	86.4%
3	74.1%	72.1%	69.2%	70.2%	89.7%
21	73.9%	79.5%	61.6%	70.7%	87.5%

D.0.4 Conv4a Layer - Top 10 Channels

Table D.4 presents the top 10 channels in Conv4a (out of 32 total). This is the final convolutional layer before the fully connected layers. Notably, even the best channel (Ch18, 68.1%) shows substantially lower performance than individual channels in earlier layers, despite whole-layer probes achieving 97.6% accuracy. This indicates that Conv4a representations are highly distributed and no single channel carries sufficient information for accurate classification.

Table D.4: Conv4a layer top 10 channels by overall accuracy (out of 32 total channels). Bold indicates highest accuracy in each column.

Channel	Overall	Blue Key	Green Key	Red Key	Gem
18	68.1%	83.3%	45.4%	70.7%	88.0%
8	67.1%	86.5%	50.0%	59.1%	73.4%
17	64.9%	83.3%	55.2%	57.7%	78.3%
28	63.0%	82.3%	51.2%	52.4%	79.9%
10	62.0%	68.8%	55.2%	58.2%	78.8%
0	61.9%	78.6%	56.4%	55.3%	66.3%
21	61.9%	71.2%	44.8%	61.5%	69.0%
4	61.8%	74.4%	47.1%	52.4%	81.5%
26	60.4%	74.0%	39.5%	51.4%	69.0%
16	59.4%	82.3%	45.4%	56.3%	64.7%

D.0.5 Summary of Channel-Level Findings

Several patterns emerge from the channel-level probe analysis:

1. **Layer hierarchy:** Conv1a contains a small number of highly informative generalist channels (Ch0, Ch11, Ch12) alongside many specialist channels that encode only specific entities. Conv2a shows improved overall performance with multiple high-performing generalist channels. Conv3a and Conv4a show increasingly distributed representations where individual channels carry less information.
2. **Distributed vs sparse coding:** The gap between whole-layer and best single-channel performance grows substantially in deeper layers (Conv3a: 99.3% whole vs 82.9% best channel; Conv4a: 97.6% whole vs 68.1% best channel), indicating a shift from sparse coding in early layers to distributed coding in later layers.
3. **Entity-specific patterns:** Some channels in Conv1a show extreme specialisation (e.g., Ch9 and Ch14 achieve 100% on red key but near 0% on other

entities), while Conv2a channels tend toward more balanced multi-entity performance.

Appendix E

Intervention Spans Across Checkpoints

Figures E.1–E.5 illustrate the representational drift observed across checkpoints 40k–60k. Despite stable task performance, the specific channels exhibiting steering behaviour and their associated activation ranges shift dramatically between checkpoints, demonstrating that the network continuously explores equivalent solutions within a broad basin of the loss landscape.

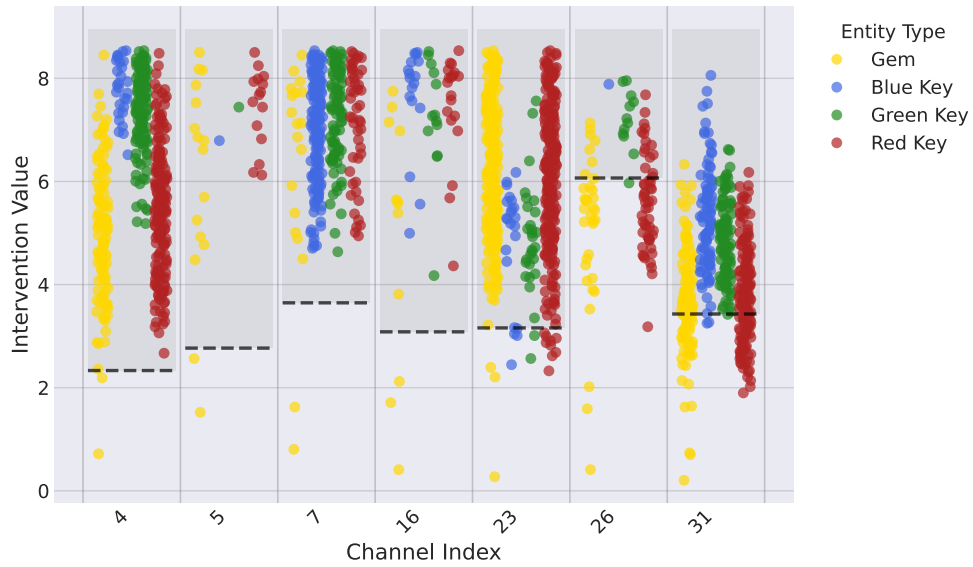


Figure E.1: Steering intervention success rates for conv4a layer at checkpoint 40k. Coloured dots represent successful steering interventions for different entities (blue key, green key, red key, gem), while black lines indicate the 99.9th percentile of normal activations. Each panel shows a different channel exhibiting steering behaviour.

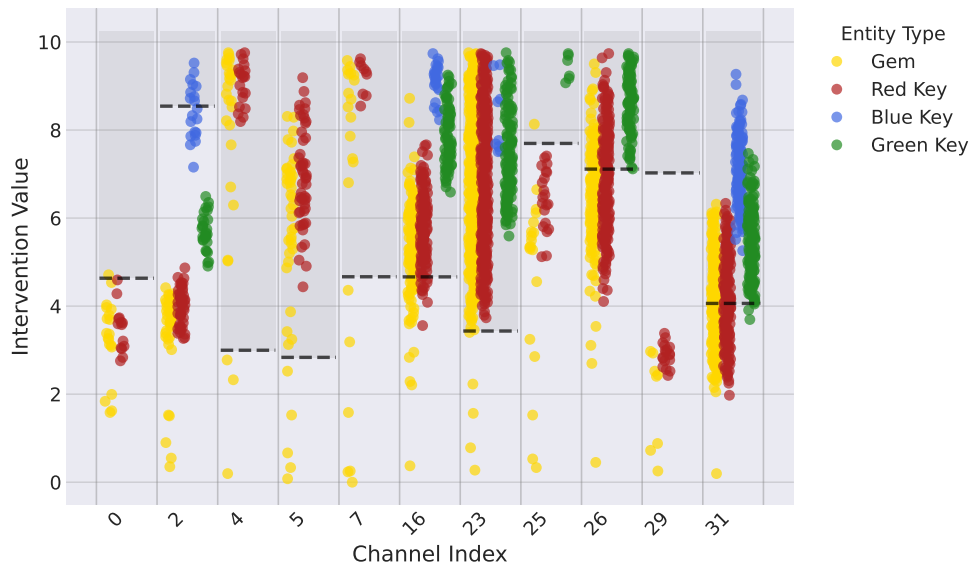


Figure E.2: Steering intervention success rates for conv4a layer at checkpoint 45k. Coloured dots represent successful steering interventions for different entities (blue key, green key, red key, gem), while black lines indicate the 99.9th percentile of normal activations. Each panel shows a different channel exhibiting steering behaviour.

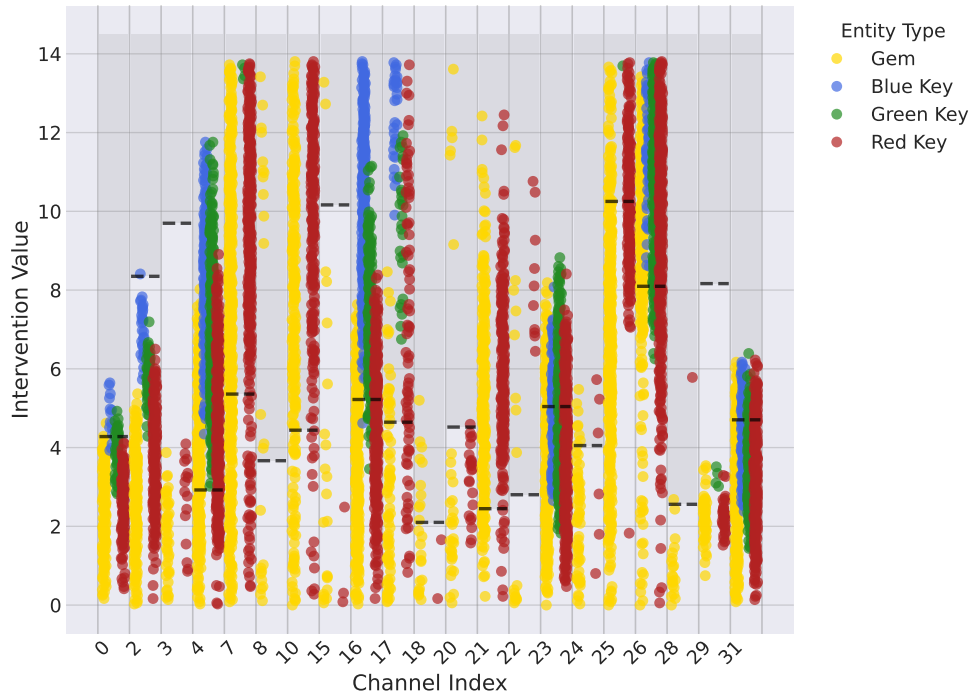


Figure E.3: Steering intervention success rates for conv4a layer at checkpoint 50k. Coloured dots represent successful steering interventions for different entities (blue key, green key, red key, gem), while black lines indicate the 99.9th percentile of normal activations. Each panel shows a different channel exhibiting steering behaviour.

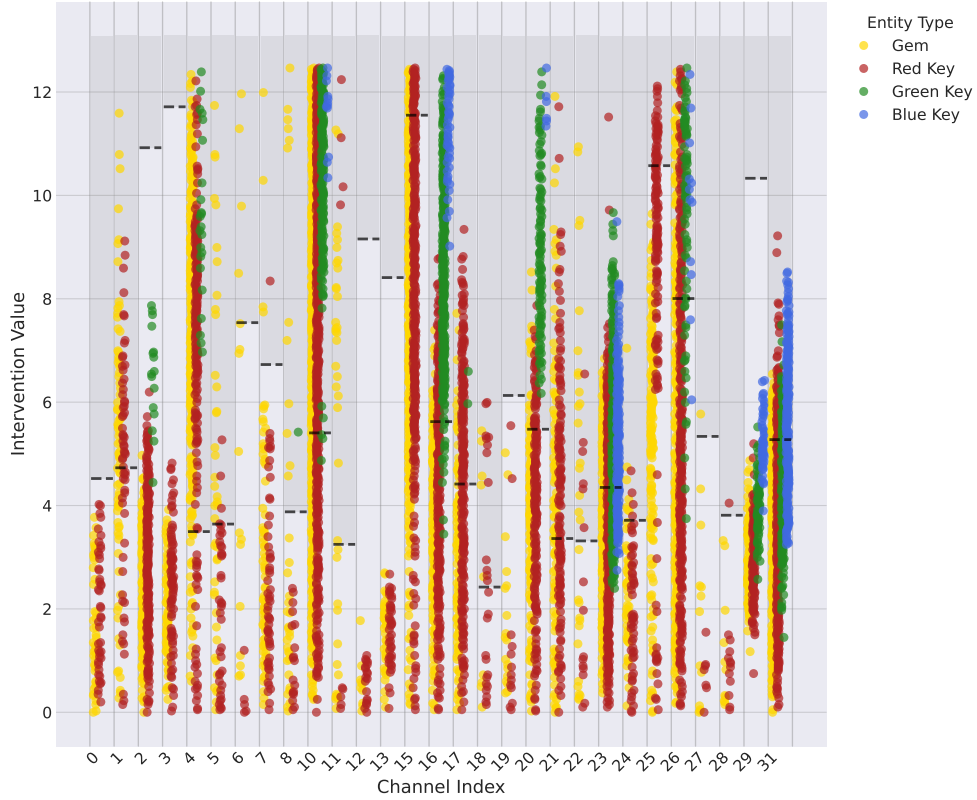


Figure E.4: Steering intervention success rates for conv4a layer at checkpoint 55k. Coloured dots represent successful steering interventions for different entities (blue key, green key, red key, gem), while black lines indicate the 99.9th percentile of normal activations. Each panel shows a different channel exhibiting steering behaviour.

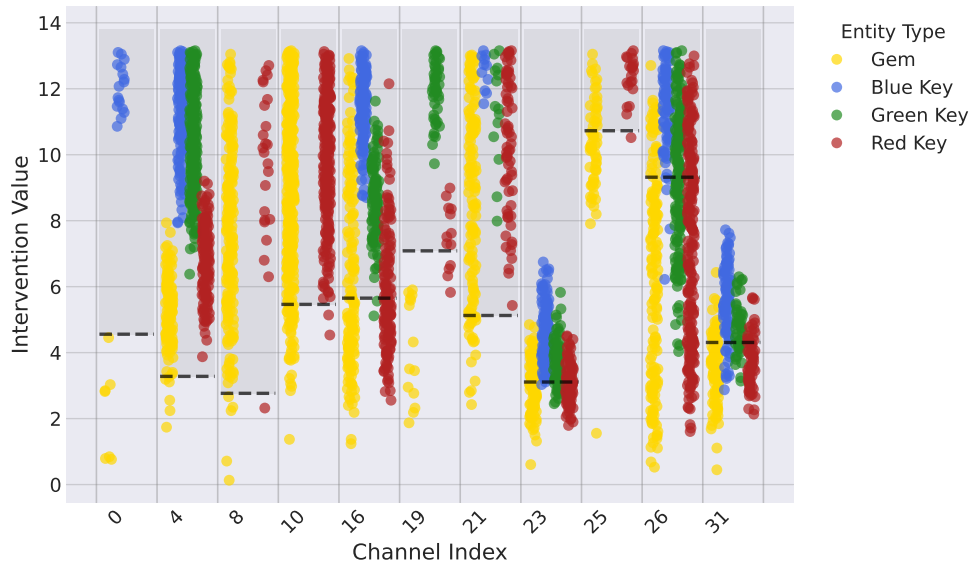


Figure E.5: Steering intervention success rates for conv4a layer at checkpoint 60k. Coloured dots represent successful steering interventions for different entities (blue key, green key, red key, gem), while black lines indicate the 99.9th percentile of normal activations. Each panel shows a different channel exhibiting steering behaviour.

Appendix F

SAE Intervention Results

The following figures show steering intervention success rates for SAE latents both trained on the 35000 step checkpoint. Unlike the base model channels (which have 32 dimensions per layer), the SAE uses a 4x expansion factor, resulting in 128 latents per layer. These figures demonstrate that sparse autoencoder latents can also be successfully steered to influence agent behaviour toward specific entities. Coloured dots represent successful steering interventions for different entities (blue key, green key, red key, gem), while black lines indicate the 99.9th percentile of normal activations during standard gameplay. Figures shown have a threshold for inclusion of ≥ 200 successes for readability.

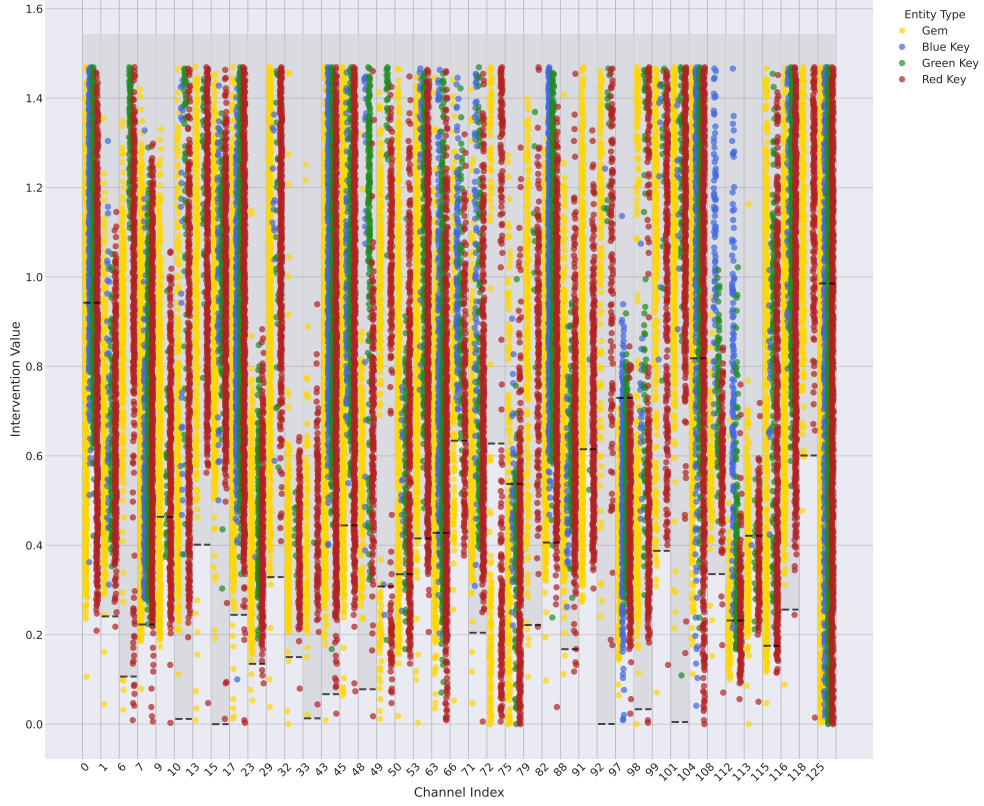


Figure F.1: SAE latent steering intervention success rates for conv3a layer, showing the top 41 most successful latents with a cutoff threshold of ≥ 200 successes. Of particular note is the large number of steering successes in general in conv3a and also the significant overlap between different sections, including within similar activation regions, displaying relatively less separation between entity activation ranges relative to elsewhere.

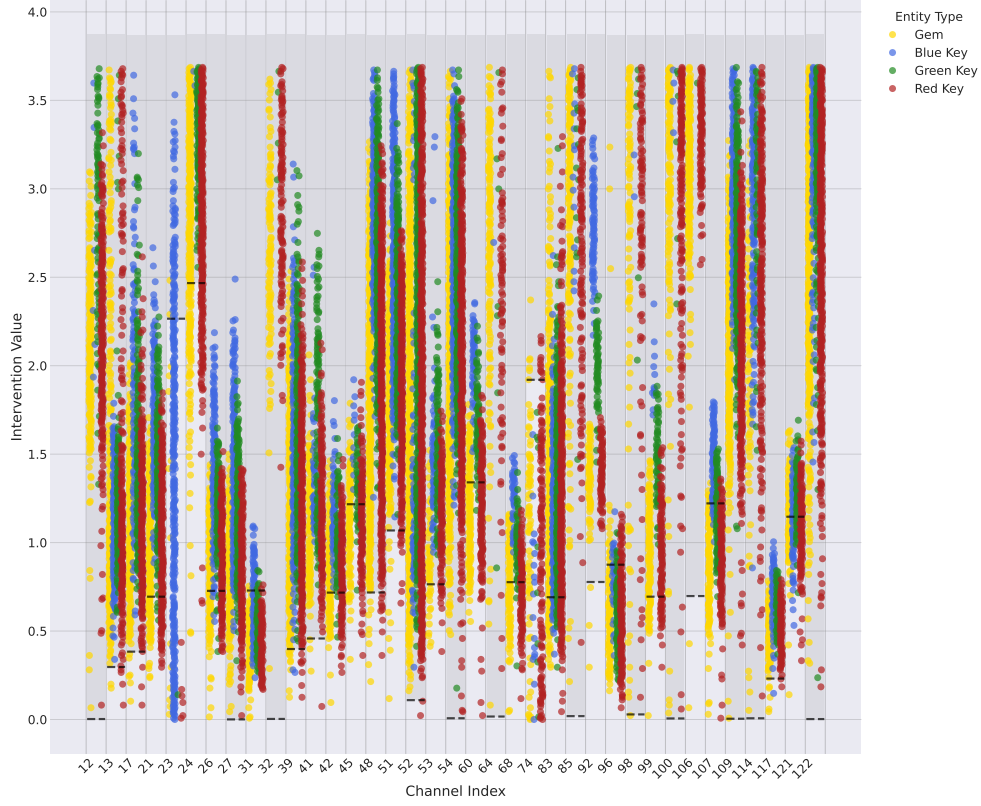


Figure F.2: SAE latent steering intervention success rates for conv4a layer, showing the top 37 most successful latents with a cutoff threshold ≥ 200 successes. Conv4a SAE shows significantly sharper separation between entity activation ranges than any other layer, in particular latent 92 shows almost perfect delineation of entities. Latent 23 also contains an unusual capacity for steering the blue key in particular, indicating that some blue key specialisation may have been learned, though we caution against making strong conclusions from the steering alone due to the complex nature of the network.

Appendix G

Expanded Ablation Studies

This appendix provides full methodology and results for the ablation experiments summarised in Section 4.6. While these experiments were extensive, the results were often counterintuitive and did not clearly illuminate the network’s core mechanisms beyond confirming that navigation is redundantly encoded.

G.1 Ablation Methodology

G.1.1 Independent Navigation Channels

To determine the role of certain channels in navigational capabilities, we developed a comprehensive ablation methodology. For each channel in the network, we ran episodes with only that channel active while masking all other channels to zero. This allowed us to measure the isolated capability of each channel to support navigation to different entities in the environment. We recorded successful entity collection counts for each channel across $n=400$ rollouts.

Because the agent has a bias towards randomly walking in a given direction depending on the current policy, for every channel a test was carried out to determine the bias direction and rotate the maze stem in this direction to avoid contamination of the results.

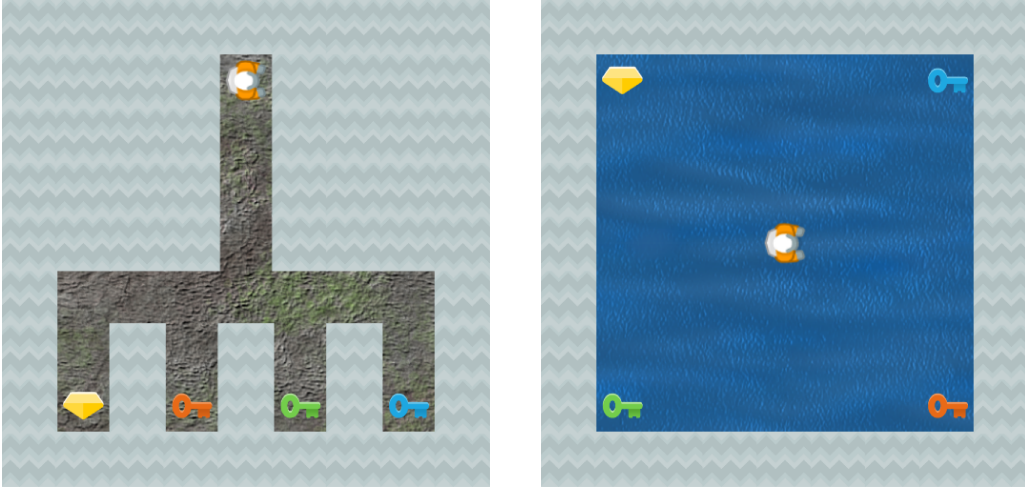


Figure G.1: Maze configurations for ablation studies. (a) Fork maze tests individual channel navigation capability by requiring explicit path choices. (b) Open maze with entities in corners tests the effect of ablating intervention spans without navigational constraints.

G.1.2 Infrastructure Channels

To establish which channels were critical to overall network function, we performed the inverse experiment: ablating only one channel at a time to observe the impact on collection rates. We repeated this for every channel in conv4a for $n=400$ roll-outs. Channels whose removal most reduced collection rates were designated as “infrastructure channels.”

G.1.3 Targeted Ablation Conditions

We performed six targeted ablations:

Ablation 1 (Intervention spans): We zero out activations falling within the ranges where patching successfully redirected the agent.

Ablation 2 (Navigation channels): We ablate the top 10 channels with highest independent collection rates.

Ablation 3 (Infrastructure channels): We ablate the top 10 channels whose removal most reduced collection rates.

Ablation 4 (Combined): We ablate both navigation channels and intervention spans simultaneously.

Ablation 5 (Inverse): We preserve only the intervention spans, ablating everything else.

Ablation 6 (Random): We randomly ablate 10 channels as a baseline.

G.2 Full Ablation Results

Table G.1: Effect of targeted ablations on entity collection rates (n=400 trials per condition). Values shown with 95% confidence intervals.

Condition	Gem	Blue Key	Green Key	Red Key
Baseline (No Ablation)	74.5% $\pm$ 4.3%	98.5% $\pm$ 1.2%	95.0% $\pm$ 2.1%	80.0% $\pm$ 3.9%
Navigation Channels	45.0% $\pm$ 4.9%	69.6% $\pm$ 4.5%	55.9% $\pm$ 4.9%	43.1% $\pm$ 4.9%
Infrastructure Channels	27.3% $\pm$ 4.4%	70.2% $\pm$ 4.5%	34.5% $\pm$ 4.7%	26.5% $\pm$ 4.3%
Intervention Spans	64.8% $\pm$ 4.7%	99.8% $\pm$ 0.4%	97.8% $\pm$ 1.4%	83.2% $\pm$ 3.7%
Navigation + Intervention	32.1% $\pm$ 4.6%	71.9% $\pm$ 4.4%	50.7% $\pm$ 4.9%	37.1% $\pm$ 4.7%
Preserve Only Intervention	31.8% $\pm$ 4.6%	61.8% $\pm$ 4.8%	43.6% $\pm$ 4.9%	32.5% $\pm$ 4.6%
Random Ablation (Control)	44.1% $\pm$ 4.9%	74.9% $\pm$ 4.3%	63.1% $\pm$ 4.7%	48.9% $\pm$ 4.9%

Table G.2: Effect of ablating top 10 infrastructure channels for each entity on collection rates (n=400 trials per condition). Values shown with 95% confidence intervals.

Condition	Gem	Blue Key	Green Key	Red Key
Baseline	74.5% $\pm$ 4.3%	98.5% $\pm$ 1.2%	95.0% $\pm$ 2.1%	80.0% $\pm$ 3.9%
Ablate Gem Infrastructure	27.3% $\pm$ 4.4%	70.2% $\pm$ 4.5%	34.5% $\pm$ 4.7%	26.5% $\pm$ 4.3%
Ablate Blue Key Infrastructure	32.5% $\pm$ 4.6%	80.2% $\pm$ 3.9%	52.8% $\pm$ 4.9%	38.0% $\pm$ 4.8%
Ablate Green Key Infrastructure	23.2% $\pm$ 4.1%	97.5% $\pm$ 1.5%	90.5% $\pm$ 2.9%	51.7% $\pm$ 4.9%
Ablate Red Key Infrastructure	49.8% $\pm$ 4.9%	55.2% $\pm$ 4.9%	49.2% $\pm$ 4.9%	46.0% $\pm$ 4.9%

G.3 Observations

The results reveal complex interactions that resist simple interpretation. Ablating gem intervention spans paradoxically *improved* collection rates (from 74% to 93.2%), likely because gem-related spans appear across 31/32 channels while key spans appear in only 14. Key intervention spans are nested within gem spans, so ablating gem regions removes key functionality along with gem signals.

Additionally, intervention signals interfere with key collection when navigation channels are disabled. Ablating navigation channels alone reduced key collection

to 45% average, but additionally ablating intervention spans partially restored performance to 53%. This suggests that intervention spans become actively harmful when navigation circuitry is impaired.

These counterintuitive results highlight the distributed and interdependent nature of the network’s representations, where individual components cannot be cleanly isolated without affecting the function of others.